

NAVFarm Application

FUNCTIONAL PROCESS DESIGN (V0.1)

Row Colour Coding

Setup / Input / Batch creation	Daily / Weekly operations	Output / Harvest / Mature stage	Mortality / Slaughter / Risk event	Variance / Close / Amortisation / Fair value	Configuration / Master setup
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Key Design:

- Every step that touches inventory, cost, or a biological asset automatically creates a double-entry journal.
- No manual GL posting is required. The system derives the entry from the entry_type_code and item_gl_mapping table.
- Standard costing batches additionally auto-create variance journals on CLOSE.
- FIFO and BIO_ASSET batches never create variance entries their true cost IS the batch cost.

Company Identity:

Field Label	Input Type	Placeholder / Options	Req?	Business Rule / Validation
Company Legal Name	Text input	For Exp- Green Valley Farms Private Limited	*	Must match Certificate of Incorporation. Max 200 characters.
Company Display Name	Text input	Short name shown in UI header	o	If blank, defaults to Company Legal Name. Max 100 chars.
Company Type	Dropdown	Select: Sole Proprietor / Partnership / Pvt Ltd / LLP / Trust / NGO / Co-operative	*	Affects tax treatment and document templates.
Industry Type	Dropdown	Agriculture / Agri-Tech / Food Processing / Agri Input / Mixed	*	Determines default module recommendations.
Registration Number	Text input	For Example- U01100MH2020PTC123456 (CIN) or equivalent	o	Company registration / CIN number.
Tax ID (GSTIN/PAN/VAT)	Text input	For Exp- 27AABCG1234F1ZK	o	Primary tax identifier. Format validated by country selection.
Incorporation Date	Date picker	DD/MM/YYYY	o	Date of legal incorporation.
Company Website	URL input	https://www.userrcompany.com	o	Shown on reports and QR code landing pages.

Support Email	Email input	support@usercompany.com	○	Used as reply-to on all outbound system emails.
Primary Phone	Phone input	+91 XXXXX XXXXX	○	E.164 format. Country code auto-filled from country selection.
Company Logo	File upload	Upload PNG or SVG. Max 2MB. Min 200×200px.	○	Shown on reports, invoices, QR frames, and mobile app.
Brand Primary Colour	Colour picker	#1F4E79	○	Hex colour used in UI, report headers, and QR code frame. Default: NAVFarm navy blue.

Company Address

Field Label	Input Type	Placeholder / Options	Req?	Business Rule / Validation
Address Line 1	Text input	Street number, building name, road	*	Max 200 characters.
Address Line 2	Text input	Area, landmark (optional)	○	Max 200 characters.
City	Text input	For Example- Pune	*	-
State / Province	Dropdown	Select from list - filtered by Country	*	Drives GST state code (India) and regional tax defaults.
Country	Dropdown	Select country	*	Determines timezone, currency, and language pre-fills.
PIN / ZIP Code	Text input	For Example- 411001	*	Format validated by country selection.
GPS Coordinates (Farm)	Map picker / Text	Latitude, Longitude - For Example- 18.520430, 73.856744	○	Used for farm location map in dashboard and weather integration.

Contacts

Field Label	Input Type	Placeholder / Options	Req?	Business Rule / Validation
Contact Type	Dropdown	Owner / CEO / CFO / Farm Manager / Accountant / IT Admin	*	-

Full Name	Text input	For Example- Rajesh Kumar Sharma	*	Max 200 characters.
Designation	Text input	For Example- Head of Farm Operations	o	Shown on correspondence.
Email Address	Email input	For Example- rajesh@userrcompany.com	*	Receives all critical system alerts and reports. Must be unique in the system.
Mobile Number	Phone input	+91 XXXXX XXXXX	o	For SMS and WhatsApp KPI alerts.
Receive KPI Alerts?	Toggle	YES / NO - default: YES	o	If YES, this contact gets immediate push/SMS for CRITICAL batch KPI violations.
Receive Weekly Reports?	Toggle	YES / NO - default: YES	o	If YES, this contact receives automated weekly batch summary and P&L report.

Language Settings

NAVFarm should supports 16 languages. English is pre selected as the default.

Field Label	Input Type	Placeholder / Options	Req?	Business Rule / Validation
Default Language	Dropdown with flag	GB English (pre-selected) change if needed	*	ALL UI labels, notifications, and PDFs will render in this language by default. Individual users can override in their profile.
Enable Additional Languages	Multi-select		o	Selected languages will appear in each user's language preference picker.
Date Display Format	Dropdown	dd/Mm/yyYY !MM/DD/YYYY !YYYY-MM-DD	*	Applies to all dates shown in reports, batch entries, and QR codes.
Number Format	Dropdown	Indian (1,00,000) !International (100,000) !European (100.000)	*	Determines lakh/crore vs million/billion grouping in financial reports.

Currency & Fiscal Yr

The base currency drives all accounting entries and cannot be changed after the first transaction is posted.

Field Label	Input Type	Placeholder / Options	Req?	Business Rule / Validation
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Base (Accounting) Currency	Dropdown		*	ALL GL journals, WIP, variance entries, and reports are in this currency. Cannot be changed after first batch transaction.
Additional Reporting Currencies	Multi-select	USD / EUR / GBP / AED / SGD/ optional	○	Financial reports can be viewed in these currencies using daily exchange rates.
Fiscal Year Start Month	Dropdown		*	Determines period-end dates for P&L, Balance Sheet, and variance reports.
Accounting Standard	Dropdown	IND AS (Indian Accounting Standards) !IFRS !US GAAP !Local GAAP	*	Determines biological asset (IAS 41) treatment and depreciation rules.
Inventory Valuation Method	Dropdown	STANDARD COSTING (recommended) !FIFO !Weighted Average	*	Company-level default. Can be overridden at LOB level for individual production batches.
GST Filing Frequency	Dropdown	Monthly !Quarterly	○	For GST return filing reminders (India only). Shown only if country = India.

Production Modules

Select the farming verticals (NOBs) user company operates. User can add more modules later from Settings. Only the modules user enable will appear in the navigation menu.

Use Module Icons

Module	Code	Description	Enable?	Includes Sub-LOBs
 Poultry	MODULE_POULTRY	Rearing, Laying, Hatching, CB Farming, Slaughter	☐ Enable	Rearing & Breeding, Laying (Std/FIFO), Hatching, CB Farming, Slaughter + QC + QR
 Livestock	MODULE_LIVESTOCK	Cattle, Piggery, Goat & Sheep, IAS 41 Bio Asset	☐ Enable	Cow/Buffalo, Piggery (Gilt+Boar+Commercial), Goat & Sheep
 Agriculture	MODULE_AGRI	Crops, Fruits, Seeds, Bearer Plant + FIFO harvests	☐ Enable	Fruit Farming (Pre+Mature), Crop Farming, Seed Processing, Flowers
 Aquaculture	MODULE_AQUA	Fish, Shrimp, Bio Asset grow-out + Slaughter	☐ Enable	Fish Farming, Aqua Slaughter + QC + QR
 Insect Farming	MODULE_INSECT	Bee Keeping, BSF (Black Soldier Fly)	☐ Enable	Bee Keeping, BSF Larvae
 Feed Production	MODULE_FEED	Feed Mill, Bill of Recipe (BOR) based batch costing	☐ Enable	Feed Production (all species), BOR management

 Quality Control	MODULE_QC	QC batch management across all LOBs	☐ Enable	QC parameter master, QC batches, pass/fail, hold management
 QR Traceability	MODULE_QR	Farm to fork QR code generation and scanning	☐ Enable	QR generation, pack management, consumer scan landing page
 Finance & GL	MODULE_FINANCE	GL, variance engine, P&L, Balance Sheet	☐ Enable	Chart of Accounts, GL journal, variance auto-calc, financial reports
 Analytics	MODULE_ANALYTICS	KPI dashboards, batch performance, trend analysis	☐ Enable	Live dashboards, historical trends, benchmark comparisons

Administrator Account

This creates the first user account with SUPER_ADMIN role. Additional users are invited after setup is complete.

Field Label	Input Type	Placeholder / Options	Req?	Business Rule / Validation
Full Name	Text input	e.g. Rajesh Kumar Sharma	*	This name appears on all actions (batch approvals, entries) this user performs.
Email Address	Email input	e.g. admin@userrcompany.com	*	This is the login username. Must be unique across the entire NAVFarm platform.
Mobile Number	Phone input	+91 XXXXX XXXXX	o	For OTP login and SMS alerts.
Password	Password input	Min 8 chars, 1 upper, 1 number, 1 special char	*	Stored as bcrypt hash. NAVFarm never sees userr plaintext password.
Confirm Password	Password input	Re-enter password	*	Must match password field exactly.
Enable 2-Factor Auth (2FA)	Toggle	YES (recommended) / NO	o	If YES, a one-time code is required on each login via Authenticator app or SMS.
Timezone	Dropdown	Auto-detected from browser / Select manually	*	Userr personal timezone for display. Defaults to company timezone.
Language	Dropdown	GB English (default) - change if needed	*	Userr personal UI language. Defaults to company default language.

What Happens After User Click 'Save & Complete Setup'

-  All data entered above is written to the database (company_master, company_address, company_contacts, company_fiscal, company_modules, user_master)
-  Userr admin account is created with SUPER_ADMIN role and user are automatically logged in
-  The system sends a welcome email to the admin email address (in the selected language)
-  Company onboarding_status is updated to COMPLETED
-  NAVFarm main dashboard opens userr selected modules appear in the navigation menu

- A setup checklist widget appears on the dashboard showing optional steps still remaining (invite team, load master data, configure GL)
- All subsequent users invited by the admin will see the company language as their default

Company Setup logic and design

Initial Setup Wizard- Step Logic

Setup wizard is a 15 step sequential onboarding flow. Steps 1 to 9 are MANDATORY, sys is locked (read only dashboard, no batch creation) until all mandatory steps are COMPLETED. Steps 10 to 14 are OPTIONAL and can be done later from Settings.

1	Company Profile	Enter company name, type, registration number, logo, brand colour, website. company_master: company_name, company_type, registration_no, company_logo_url, primary_color_hex	✓ MANDATORY 5 min	
2	Address	Enter registered office address and primary farm GPS location. company_address: line1, city, state_id, country_id, pincode, gps_latitude, gps_longitude	✓ MANDATORY 3 min	
3	Key Contacts	Add primary contact (owner or manager) name, email, phone, and alert preferences. company_contacts: full_name, email, phone_primary, receives_alerts, receives_reports	✓ MANDATORY 5 min	
4	Default Language LANGUAGE	Select company default language. English is pre-selected. Enable additional languages for team. company_master: default_language_id. company_language_config: lang_id, is_default, is_enabled	✓ MANDATORY 2 min	
5	Base Currency CURRENCY	Select accounting base currency. INR pre-selected for India. Cannot change after first transaction.	✓ MANDATORY 2 min	

		company_master: base_currency_id. company_currency_config: is_base=TRUE		
6	Timezone & Region TIMEZONE	Select company timezone (auto-detected from browser). Drives all batch scheduling and timestamps. company_master: default_timezone_id, country_id	✓ MANDATORY 2 min	
7	Fiscal Year & Accounting FISCAL	Select fiscal year start month, accounting standard (IND AS/IFRS), inventory valuation, period type. company_fiscal: fiscal_year_format, fiscal_start_month, accounting_standard, inventory_valuation	✓ MANDATORY	
8	Enable Modules MODULES	Check which NOB/LOB modules to activate. Drives navigation, schedulers, and GL accounts loaded. company_modules: module_code, is_active, activated_on	✓ MANDATORY	
9	Admin User Account USERS	Set first user: name, email, password, 2FA, timezone, language. Auto-assigned SUPER_ADMIN role. user_master + user_role_assignment (role=SUPER_ADMIN)	✓ MANDATORY	
10	Invite Team Members USERS	Send email invites to farm managers, accountants, supervisors. Assign roles on invite. user_master: email, user_type. Invitation email sent in company default language.	○ OPTIONAL	
11	Chart of Accounts GL	Review and confirm the pre-loaded GL COA. Add company-specific accounts if needed. gl_account_master: account seeded based on accounting_standard + enabled modules.	✓ MANDATORY	

12	NOB & LOB Config NOB_LOB	For each enabled module: confirm costing method, scheduler templates, KPI thresholds. nob_master, lob_master, scheduler_parameter_master seeded from module defaults.	✓ MANDATRY	
13	Master Data Load MASTER_DATA	Load or confirm: UOM, Item, Breed, Location, Resource masters. System seeds defaults. uom_master, item_master, breed_master, location_master, resource_master	○ OPTIONAL	
14	Notification Settings NOTIFICATIONS	Configure email SMTP, SMS gateway, Push FCM key. Send test notifications. notification_config: smtp_host, sms_provider, push_fcm_key_enc	○ OPTIONAL	
15	Setup Complete COMPLETE	Wizard closed. Dashboard unlocked. onboarding_status → COMPLETED. company_master: onboarding_status = COMPLETED. Welcome dashboard shown.	○ OPTIONAL	

Multi Language- Runtime Resolution Logic

NAVFarm uses a 3 level language resolution chain. At runtime, every UI label, notification and report string is resolved using this priority order:

Priority	Source	How it works	When it applies
1 – HIGHEST	User preference (user_language_pref)	If the logged-in user has set a personal language preference, ALL UI strings render in that language.	When user has explicitly changed their language in My Profile → Language.
2 – MIDDLE	Company default (company_master.default_language_id)	If no user preference exists, all UI strings render in the company's configured default language.	All users who have not set a personal preference.
3 - FALLBACK	System default (English - is_system_default=TRUE)	If a translation key is missing in the selected language (coverage <	For any language with translation_coverage_pct <

		100%), the English string is shown instead of a blank.	100%. Prevents blank labels.
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Where Language Applies in NAVFarm

- UI Labels & Menus: All button text, field labels, column headers, menu items, error messages.
- Notifications: KPI alert emails, SMS, push notifications sent in the recipient's language preference.
- PDF Reports: Batch summary, P&L statement, variance report printed in the company default language.
- QR Code Landing Page: The consumer facing page shown when a QR code is scanned uses company default.
- Validation Messages: Form error messages (for example- This field is required) shown in users language.
- Sys Emails: Welcome email, password reset, invitation sent in company default language.
- Admin Exception: SUPER_ADMIN always sees English for system-level config pages (e.g. GL setup, server config).

User System Roles

Role	Can Create Batches	Can Approve	Can View Cost	Can Access Finance	Can Admin Users	Can Export	Notes
SUPER_ADMIN	YES	YES	YES	YES	YES	YES	Full access. One per company (first setup user). Cannot be modified.
ADMIN	YES	YES	YES	YES	YES	YES	Same as Super Admin except cannot delete the Super Admin user.
FARM_MANAGER	YES	YES	YES- own farms	NO	NO	YES	Can create, edit, and approve batches for their assigned farm locations.
ACCOUNTANT	NO	NO	YES - all	YES	NO	YES	Read-only on batches. Full access to GL, variance, and financial reports.
AUDITOR	NO	NO	YES - all	YES - read only	NO	YES	Read-only across all modules. Can export. Cannot create or modify.
SUPERVISOR	YES	NO	YES- own batches	NO	NO	NO	Daily batch data entry. Cannot approve. Cannot

							see batch cost totals.
VIEWER	NO	NO	NO	NO	NO	NO	Dashboard and KPI view only. No data entry.
CUSTOM	configurable	configurable	configurable	configurable	configurable	configurable	Admin creates custom roles with granular permission matrix.

Sheet 08- Poultry LOBs: Step by Step Proces

LOB: Rearing & Breeding

Batch: BATCH-REAR-2025-001 ! 5,000 DOC ! 35 Days ! Method: STANDARD

This is the first link in the poultry chain. Day Old Chicks are placed into a rearing shed. They are fed and managed until they become grower chicks, which are then either transferred to a laying batch or CB farming batch.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr --> Cr)	Business Impact & Outcome
S1	SETUP	System pre-loads the standard cost for Starter Feed into standard_cost_master. This locked rate is the benchmark all consumption will be measured against.	= INSERT standard_cost_master: item=FEED_STARTER, lob=PLT_REARING std_cost = 18.00 Rs./KG	No journal (master data)	Establishes the cost benchmark. Without this, no variance can be calculated at batch close. Must be done before batch is APPROVED.
S2	SETUP	System pre-loads the standard cost for Grower Feed. Both starter and grower rates are locked at batch APPROVE - they cannot change mid-batch.	= INSERT standard_cost_master: item=FEED_GROWER std_cost = 20.00 Rs./KG	No journal (master data)	Grower feed typically has higher protein. Locking the rate enables price variance calculation when actual purchase price differs.
1	DRAFT→APPROVED	User creates the batch header and places 5,000 DOC. Application: Creates a batch_input_lines record linking to the source inventory lot.	= qty × std_unit_cost = 5,000 NOS × Rs.22.00 = Rs.1,10,000 charged to WIP	Dr 1190 WIP → Cr 1110 Inv.Birds Rs.1,10,000	Batch is now live and costing has begun. Any DOC mortality from this moment onward is tracked against this opening balance. Sys validates that the shed location has capacity ≥ 5,000 birds.

		- Resolves and locks the costing method as STANDARD. - Moves batch status from DRAFT to APPROVED. - Issues the DOC from Bird Inventory into the batch WIP account.			
2	DAY 1	Daily feed entry: farm worker records actual feed consumed. App: - Reduces Feed Inventory by the actual quantity. - Adds that cost to Batch WIP. - Compares actual vs expected qty and stores the difference for variance calculation at close.	= expected = $0.025 \text{ kg/bird} \times 5,000 \times 1 \text{ day}$ = 125 KG expected actual = 120 KG difference = -5 KG (FAV) cost = $120 \times \text{Rs.18} = \text{Rs.2,160}$	Dr 1190 WIP → Cr 1100 Inv.Feed Rs.2,160	5 Kg shortfall is tracked but does not create a journal yet, it accumulates across all days and is settled at batch CLOSE as a usage variance. Farm manager can see running FCR daily.
3	DAY 7	Mortality event: 5 male chicks found dead. Farm worker records MORT_M entry. System: - Removes 5 birds from batch headcount. - Removes the cost of those 5 birds from WIP at their running average cost. - Posts the loss to the Mortality Loss expense account.	= qty × running_avg_cost = $5 \text{ NOS} \times \text{Rs.22.00} = \text{Rs.110}$ removed from WIP	Dr 7120 Mortality Loss → Cr 1190 WIP Rs.110	Mortality is expensed immediately - it does not inflate the cost of surviving birds. The running batch cost per bird automatically improves (same total cost spread over fewer heads). KPI alert fires if cumulative mortality exceeds the scheduler threshold (For Example- >2%).
4	DAY 35	Batch completes. Farm worker records output: 4,940 grower chicks are	= output_qty × std_output_cost =	Dr 1110 Inv.Birds → Cr	Grower chicks are now in inventory at Rs.85/bird and can be consumed by a Laying

		transferred to Bird Inventory at standard cost. Sys: - Credits WIP with the value of output (at std cost). - Debits Bird Inventory. - Remaining WIP balance (if any) represents variances not yet closed.	$4,940 \times \text{Rs.}85.00 = \text{Rs.}4,19,900$ leaves WIP	1190 WIP Rs.4,19,900	or CB batch. The batch status changes to COMPLETED pending variance settlement.
5	CLOSE	System auto calculates Usage Variance for feed. All daily feed consumption deviations are summed and one variance journal is created.	$= (\text{actual_total} - \text{std_total}) \times \text{std_rate}$ $\text{std_total} = 0.025 \times 5,000 \times 35 = 4,375 \text{ KG}$ $\text{actual} = 6,775 \text{ KG (example)}$ $\text{var} = 2,400 \text{ extra} \times \text{Rs.}18 = \text{Rs.}43,200$ PLANNED	Dr 6120 Usage Var → Cr 1190 WIP Rs.43,200 (if PLANNED)	Feed usage variance tells the farm manager whether the flock consumed more or less feed than the standard programme expected. Persistently expected usage variance signals FCR problems, feed wastage, or incorrect daily recording.
6	CLOSE	System auto calculates Output Variance. Expected output is derived from opening qty adjusted for expected mortality percentage.	$= (\text{std_expected_output} - \text{actual_output}) \times \text{std_output_cost}$ $\text{expected} = 5,000 \times 99\% = 4,975$ $\text{actual} = 4,940$ $\text{var} = 35 \times \text{Rs.}85 = \text{Rs.}2,975$ PLANNED	Dr 6140 Output Var → Cr 1190 WIP Rs.2,975 (PLANNED)	Output variance captures whether the flock produced fewer chicks than the standard programme expected. After all variances, WIP balance should be zero. If not, the system rejects the close and prompts the user to investigate.

LOB: Laying- Standard Costing

Batch: BATCH-LAY-STD-2025-001 ! 8,000 Hens ! 52 Weeks ! Method: STANDARD ! Traceable from BATCH-REAR-2025-001

Laying hens transferred from the Rearing batch produce eggs over 52 weeks. Daily feed consumption and egg output are recorded. At batch close, price variance (due to feed price changes) and usage variance (excess feed) are auto calculated.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
L1	BATCH HEADER	User creates the Laying batch. The input line explicitly references the source Rearing batch	$= \text{batch_input_lines:}$ $\text{source_batch_id} = \text{BATCH-REAR-2025-001}$ $\text{item} = \text{LAYING_HEN},$	No additional journal - cost carried from Rearing batch	Full traceability is established from egg → hen → DOC → hatchery. This is essential for farm-to-fork QR code

		(BATCH-REAR-2025-001), establishing traceability. The system copies the hen inventory into this batch's opening balance.	qty = 8,000 unit_cost = Rs.85 (from rearing batch)		generation and biosecurity incident tracing.
L2	DAY 1	Daily feed entry: 1,200 KG Layer Mash consumed. The system computes expected qty (0.150 kg/hen/day × 8,000 = 1,200 KG), records actual = expected, and adds Rs.23,400 to WIP.	= expected = 0.150 × 8,000 × 1 = 1,200 KG actual = 1,200 KG cost = 1,200 × Rs.19.50 = Rs.23,400	Dr 1190 WIP → Cr 1100 Inv.Feed Rs.23,400	Running batch cost increases by Rs.23,400/day. Running cost per egg = total_batch_cost / cumulative_eggs. Farm manager watches this daily to detect feed overuse before end of cycle.
L3	DAY 1	Egg collection recorded: 6,800 eggs enter Egg Inventory at current running cost per egg.	= lay_rate = 6,800 / 8,000 = 85% unit_cost = running_batch_cost / cumulative_eggs	Dr 1120 Inv.Eggs → Cr 1190 WIP (at running cost)	Eggs are now saleable/transferable inventory. The lay rate KPI (85%) is within the 80% minimum threshold - no alert fires. If lay rate drops below 80%, a KPI_DEVIATION alert is sent to the Farm Manager.
L4	DAY 1	Descriptive observation: lay rate 85% is recorded as a non-costing data point. This drives the KPI dashboard and historical performance tracking.	= entry_type = DESC_LAY_RATE value = 85% kpi_min = 80% → 85% ≥ 80% ✓ NO alert fired	No journal (descriptive only)	Even though no cost flows, this entry feeds the KPI engine. Consecutive days below 80% would escalate to CRITICAL. Management uses this to decide vaccination, feed adjustment, or flock replacement.
L5	DAY 7	8 hens die. MORT_F entry reduces headcount and removes cost from WIP.	= 8 NOS × Rs.35 running_avg_cost = Rs.280 removed from WIP	Dr 7120 Mortality Loss → Cr 1190 WIP Rs.280	Mortality is expensed immediately. Remaining 7,992 hens continue the batch. Running cost per egg is unaffected (both cost numerator and egg denominator adjust proportionally). Cumulative mortality KPI is updated.
L6	WEEK 52	Batch CLOSE: Price Variance calculated. Feed was purchased at Rs.20.50/KG vs the Rs.19.50/KG standard - Rs.1.00 price overrun on every KG consumed.	= (actual_rate – std_rate) × actual_qty_consumed = (20.50 – 19.50) × 86,400 KG = Rs.86,400 PLANNED	Dr 6100 Price Var → Cr 1100 Inv.Feed Rs.86,400	Price variance is procurement's responsibility, not farm operations. This journal surfaces the cost overrun from rising feed prices and enables management to hold procurement accountable or adjust the standard cost for the next cycle.
L7	WEEK 52	Batch CLOSE: Usage Variance calculated. 2,160 KG of feed was	= (actual_qty – std_qty) × std_rate std = 0.150 × 7,992 avg hens × 365 days extra = 2,160 KG	Dr 6120 Usage Var → Cr 1190 WIP Rs.42,120	Usage variance is farm operations' responsibility. 2,160 extra KG suggests poor FCR, feed spillage, or incorrect daily measurement. Combined with the price variance,

		consumed beyond the standard schedule.	$\times \text{Rs.}19.50 = \text{Rs.}42,120$ PLANNED		the batch ended Rs.1,28,520 over standard - prompting a management review.
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LOB: Laying- FIFO Costing Variant

Batch: BATCH-LAY-FIFO-2025-001 ! 8,000 Hens ! 52 Weeks ! Method: FIFO

Same laying operation as above, but using FIFO costing. Each feed receipt creates a cost layer. Consumption draws down the oldest layer first. No variance entries are generated - the true chronological cost IS the batch cost.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
F1	GRN	First feed delivery received and booked into inventory. FIFO Layer 1 is created in lot_serial_master with the receipt date and rate.	= FIFO Layer 1 created: item=FEED_LAYER qty=5,000 KG @ Rs.19.50 receipt_date=Jan 5 layer_status=OPEN	Dr 1100 Inv.Feed → Cr 2100 AP $5,000 \times \text{Rs.}19.50 =$ Rs.97,500	Layer 1 is now available for consumption. The system knows exactly which supplier delivery this feed came from, enabling recall tracing if a contaminated batch is discovered.
F2	GRN	Second delivery received at a higher price. FIFO Layer 2 is created.	= FIFO Layer 2 created: qty=3,000 KG @ Rs.21.00 receipt_date=Jan 15 layer_status=OPEN	Dr 1100 Inv.Feed → Cr 2100 AP $3,000 \times \text{Rs.}21.00 =$ Rs.63,000	Two layers now exist. The system will always consume Layer 1 first before touching Layer 2. This ensures the oldest, cheapest stock is used first - both for cost accuracy and for feed freshness.
F3	DAY 1-4	Days 1- 4: 1,200 KG consumed daily. All 4,800 KG drawn from Layer 1 only. Layer 1 reduces from 5,000 to 200 KG remaining.	= 4 days $\times$ 1,200 KG = 4,800 KG all from Layer 1 @ Rs.19.50 cost = $4,800 \times \text{Rs.}19.50$ = Rs.93,600 Layer 1 remaining = 200 KG	Dr 1190 WIP → Cr 1100 Inv.Feed Rs.93,600 (over 4 days)	Feed cost in WIP is precisely Rs.19.50/KG for these 4 days. The batch running cost per egg accurately reflects the actual purchase price - no averaging distortion. This is more precise than Standard costing for volatile commodity prices.
F4	DAY 5 LAYER SWITCH	Day 5 requires 1,200 KG but Layer 1 only has 200 Kg left. System automatically splits the consumption across both layers, closing Layer 1.	= Layer 1: 200 KG $\times$ Rs.19.50 = Rs.3,900 Layer 2: 1,000 KG $\times$ Rs.21.00 = Rs.21,000 total = Rs.24,900 blended_cost = $\text{Rs.}24,900 / 1,200 =$ Rs.20.75/KG	Dr 1190 WIP → Cr 1100 Inv.Feed Rs.24,900 Layer 1 STATUS → CLOSED	The blended rate (Rs.20.75) correctly reflects the mixed sourcing on this day. Layer 1 is permanently closed. This automatic layer switching is the core FIFO logic, sys handles it without any user intervention.
F5	DAY 6+	All subsequent consumption comes from Layer 2 at	= 1,200 KG/day $\times$ Rs.21.00 = Rs.25,200/day from Layer 2	Dr 1190 WIP → Cr 1100 Inv.Feed Rs.25,200/day	Higher unit cost (Rs.21.00 vs Rs.19.50) is reflected in the running batch cost per egg from Day 6 onward. Management can see

		Rs.21.00/KG until it is exhausted.			exactly when the more expensive feed started impacting egg cost - something Standard costing cannot show.
F6	CLOSE	FIFO batch close: NO variance entries are created. The actual FIFO layered cost is the true, auditable cost of the batch.	= No variance journal total_batch_cost = SUM(all FIFO layer consumptions) cost_per_egg = total / total_eggs	No variance entries WIP already reflects true cost	This is FIFO key advantage will be: zero variance distortion. Cost per egg perfectly matches the actual feed prices paid on each day. Preferred by auditors and for tax purposes when input prices fluctuate significantly.

LOB: Hatching

Batch: BATCH-HATCH-2025-001 ! 50,000 Eggs set ! 21 Days ! Method: STANDARD

Hatching LOB covers the complete 21-day incubation cycle from egg setting to Day-Old Chick production. In hatcheries this is a 2 machine process:

eggs spend Days 1-18 in a Setter machine and are transferred on Day 18 to a Hatcher machine for the final hatching phase. Candling inspections on Days 7 and 14 identify and remove infertile and dead in shell eggs before they reach the hatcher improving hatchability, reducing contamination risk and controlling costs.

✚ Its critical and typical process, so describing in details:

Hatchery Process:

Stage	Days	Machine	What Happens	Temperature	Humidity (For Example)
Setting	Day 1	Setter (Incubator)	Fertile eggs placed in setter trays. Eggs turned automatically every 1-2 hours to prevent embryo adhesion.	37.8°C (100°F)	50-55% RH
Early Incubation	Day 1-7	Setter	Embryo develops. Blood vessels visible on candling from Day 5.	37.8°C	50-55% RH
1st Candling	Day 7	Setter (pulled out momentarily)	Eggs candled with strong light. Infertile eggs (clear) and early dead embryos removed.	-	-
Mid Incubation	Day 8-14	Setter	Embryo grows rapidly. Air cell enlarges. Turning continues.	37.8°C	50-55% RH
2nd Candling	Day 14	Setter (pulled out momentarily)	Dead embryos (dark, no movement) removed. Remaining confirmed fertile.	-	-

Late Incubation	Day 15-18	Setter	Embryo fills most of egg. Turns continue. Egg weight loss monitored.	37.8°C	50-55% RH
Setter → Hatcher Transfer	Day 18	Transfer	Eggs moved to hatcher trays. Turning STOPS. Eggs laid flat. No more candling.	-	-
Pipping & Hatching	Day 19-21	Hatcher	Chick pips shell (internal pip Day 19, external pip Day 20). Full hatch Day 21.	37.2°C (99°F)	65-70% RH
Chick Pull	Day 21	Hatcher	Live chicks pulled, counted, sorted, vaccinated. Infertile/DIS eggs counted as by-product.	-	-

NAVFarm Batch Structure:

Element	Value / Rule
Batch Code	BATCH-HATCH-2025-001
Batch Type	PRODUCTION (sub-type: HATCHING)
Opening Input	50,000 fertile eggs from 2 source batches (multi-source input lines)
Batch Duration	21 days (Day 1 = Setting day, Day 21 = Hatch day)
Costing Methods Supported	STANDARD and FIFO (user selects)
Sub-Locations	Setter Room (Days 1-18) → Hatcher Room (Days 18-21)
Candling Events	Day 7 (1st candling) and Day 14 (2nd candling) - both create CANDLING batch entries
Output 1 - Main	DOC (Day-Old Chick) - live, healthy chicks
Output 2 - By-product	Infertile eggs (never developed) + DIS eggs (dead-in-shell) - sold as waste/animal feed
Output 3 - Waste	Broken shells, unusable material - discarded
KPI - Hatchability	$\text{hatchability} = \text{live_chicks} / \text{total_eggs_set} \times 100$. Target $\geq 87\%$
KPI - Fertility Rate	$\text{fertility} = (\text{total_set} - \text{infertile_candled}) / \text{total_set} \times 100$. Target $\geq 92\%$
KPI - Candling Loss Rate	$\text{candling_loss} = (\text{infertile} + \text{DIS}) / \text{total_set} \times 100$. Alert if $> 10\%$
GL WIP Account	1190 Work-In-Progress (Batch WIP)
GL Output Account	1110 Inventory – Live Birds (DOC chicks)
GL By-product Account	1100 Inventory – Feed & Consumables (infertile eggs sold as feed/fertiliser)

Batch- STANDARD Costing Method

Batch: BATCH-HATCH-2025-001 !50,000 Eggs Set !Method: STANDARD

All numbers used throughout this document. The Standard Method uses a locked standard cost per chick (Rs.4.20). Variances are auto-generated at batch close.

Item	Value	Comments
Total eggs set	50,000 NOS	Input to setter on Day 1
Source 1	30,000 eggs from BATCH-LAY-2025-001	Laying batch - commercial layer hens
Source 2	20,000 eggs from BATCH-BREED-2025-001	Breeder flock - pure-line genetics
Egg cost (std)	Rs.3.20 per egg	Standard cost from laying/breeding batch
Total opening WIP	Rs.1,60,000	50,000 × Rs.3.20
Day 7 Candling Removal	2,500 infertile eggs	5% of set = clear/infertile
Day 14 Candling Removal	1,000 DIS eggs	2% of remaining = dead-in-shell
Remaining in hatcher	46,500 eggs	Set for final hatch
Standard hatchability	87% of eggs set	Industry standard for healthy flock
Expected live chicks	43,500	50,000 × 87%
Actual live chicks	42,500	Actual count on Day 21
Actual hatchability	85%	42,500 / 50,000
DIS in hatcher (unhatched)	4,000	46,500 – 42,500
Total non-hatch eggs	7,500	2,500 (Day7) + 1,000 (Day14) + 4,000 (DIS hatcher)
Incubator electricity (21 days)	Rs.8,500	OVHD_POWER
Labour - setter loading	Rs.1,200	Setting labour Day 1
Labour - candling (Day 7)	Rs.600	Candling team labour
Labour - candling (Day 14)	Rs.600	Candling team labour
Labour - hatcher transfer	Rs.800	Transfer team Day 18

Labour - chick pull & sort	Rs.1,500	Chick pull team Day 21
Vaccine (Marek's Day 21)	Rs.2,125	42,500 doses × Rs.0.05/dose
Total overhead + labour	Rs.15,325	All incubation overhead
Total batch cost	Rs.1,75,325	Rs.1,60,000 eggs + Rs.15,325 overhead
Standard output cost/chick	Rs.4.20	Std cost locked at batch APPROVE
Std total output value	Rs.1,82,700	43,500 std chicks × Rs.4.20
By-product NRV (infertile eggs)	Rs.0.50/egg	Net Realisable Value for waste eggs
By-product value	Rs.3,750	7,500 × Rs.0.50
Cost split - main/by-product	95% / 5%	Main product absorbs 95% of WIP cost

Costing calculation- Day by Day Process

Batch: BATCH-HATCH-2025-001 !Method: STANDARD !50,000 Eggs !21 Days

Standard costing locks rates at batch APPROVE. Variances (output variance and overhead variance) are auto-generated at batch CLOSE. Every physical event - egg setting, candling, machine transfer, hatch - creates a system entry.

Row Colour Guide

SETUP / Batch Header	Setter Stage / GRN / FIFO	Candling Inspection	Setter→Hatcher Transfer	Hatch Output / Live Chicks	Infertile / Dead / Waste	Close / Variance	Daily Overhead / Power
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Batch Setup & Egg Setting (Day 1)

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
S1	Pre-Day 1	SETUP	System pre-loads standard cost for DOC chick output: Rs.4.20/chick. This is locked at batch APPROVE and cannot change during the cycle.	= INSERT standard_cost_master: item = DOC_CHICK lob = PLT_HATCHING std_cost = 4.20 Rs./NOS	No journal (master data)	Without the standard cost, no output variance can be computed at close. Must be configured before any hatching batch is approved.
S2	Pre-Day 1	SETUP	Standard overhead rate per egg set: Rs.0.306/egg (Rs.15,325 total overhead / 50,000	= std_oh_per_egg = total_std_overhead / opening_qty = Rs.15,325 / 50,000 = Rs.0.3065/egg	No journal (master data)	The overhead rate per egg is the absorption base for overhead variance calculation at batch close.

			eggs). Locked at batch APPROVE.			
H1a	Day 1	BATCH HEADER	User creates the hatching batch header. Two input lines are created - one for each source batch - each with a source_batch_id. The system resolves and locks costing method as STANDARD. Sub-location set to Setter Room.	= batch_master: batch_code=BATCH-HATCH-2025-001 lob_id = PLT_HATCHING opening_qty = 50,000 sub_location = SETTER_ROOM resolved_method = STANDARD (LOCKED)	No journal (batch creation)	Batch is now DRAFT. No cost flows until APPROVE. Sub-location = Setter Room means all eggs are physically in the setter machine. Capacity of setter is validated - must be ≥ 50,000 egg positions.
H1b	Day 1	EGG INPUT - Source 1	30,000 eggs drawn from Laying batch (BATCH-LAY-2025-001) inventory. Input line 1 created with source_batch_id for full traceability. Batch status → APPROVED.	= batch_input_lines Line 1: source_batch_id=BATCH-LAY-2025-001 item = HATCH_EGG qty = 30,000 NOS unit_cost = Rs.3.20 total = Rs.96,000	Dr 1190 WIP → Cr 1100 Inv.Eggs Rs.96,000	30,000 eggs leave Egg Inventory and enter Hatching WIP. Traceability established: these chicks (if hatched) can be traced back to the specific laying flock and even the laying batch date.
H1c	Day 1	EGG INPUT - Source 2	20,000 eggs drawn from Breeder batch (BATCH-BREED-2025-001). Input line 2 created.	= batch_input_lines Line 2: source_batch_id=BATCH-BREED-2025-001 item = HATCH_EGG qty = 20,000 NOS unit_cost = Rs.3.20 total = Rs.64,000	Dr 1190 WIP → Cr 1100 Inv.Eggs Rs.64,000	Total opening WIP = Rs.96,000 + Rs.64,000 = Rs.1,60,000. The batch now has 50,000 eggs in WIP from 2 source batches. If a disease is found in chicks later, the system can identify which source batch the affected eggs came from.
H1d	Day 1	SETTER LOADING LABOUR	Setter loading labour Rs.1,200 recorded as overhead entry. Workers physically place eggs onto setter trays in correct orientation (pointed end down).	= entry_type = OVHD_LABOR desc = 'Setter Loading Labour' amount = Rs.1,200 added to WIP overhead bucket	Dr 1190 WIP → Cr 2100 AP Rs.1,200	Labour for setter loading is a direct incubation cost. The Rs.1,200 becomes part of the DOC unit cost. Tracking this separately enables benchmarking: loading time per 1,000 eggs helps capacity and staffing planning.

Setter Stage- Daily Overheads (Days 1 to 8)

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
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H2a	Day 1-18	OVHD_POWER (Daily)	Incubator electricity recorded daily as an overhead entry. The setter runs 24 hours/day maintaining 37.8°C and 50-55% humidity. Daily power cost = Rs.8,500 total / 21 days ≈ Rs.405/day.	= daily_power_cost = Rs.405/day total 18 setter days: Rs.405 × 18 = Rs.7,290 (setter share) entry_type = OVHD_POWER	Dr 1190 WIP → Cr 2100 AP Rs.405/day	Electricity is the largest running cost in a hatchery. Monitoring daily power spend vs the standard Rs.405/day catches compressor faults, door seal failures, or generator use events. A KPI_DEVIATION alert fires if daily power exceeds Rs.486 (120% threshold).
H2b	Day 1-18	DESC_TEMP (Daily KPI)	Setter temperature recorded twice daily as a descriptive observation entry. KPI: 37.8°C (±0.2°C). No cost impact - pure KPI monitoring.	= entry_type = DESC_TEMP value = 37.8°C kpi_min = 37.6°C kpi_max = 38.0°C deviation → alert if outside range	No journal (descriptive KPI)	A temperature spike above 38.0°C for more than 2 hours permanently damages embryos - no recovery. A CRITICAL alert is sent immediately to the Hatchery Manager's mobile phone. This is the most important real-time KPI in the entire hatching process.
H2c	Day 1-18	DESC_HUMIDITY (Daily KPI)	Setter humidity recorded. Target 50-55% RH. Humidity below 45% causes excessive egg weight loss (dehydrated embryo). Above 60% causes drowning.	= entry_type = DESC_HUMIDITY value = 52% RH kpi_min = 50% kpi_max = 55% → WARNING alert if outside range	No journal (descriptive KPI)	Egg weight loss target over 18 setter days is 12 to 13% of initial egg weight. If humidity is too high, weight loss is less - chick drowns at hatch. Too low - chick dehydrates. Daily humidity logging enables corrective action before hatch quality is affected.
H2d	Day 1-18	DESC_EGG_TURNING (Daily)	Setter turns eggs automatically every 2 hours. System logs a confirmation entry that turning is active. If turning motor fails, a CRITICAL alert fires.	= entry_type = DESC_EGG_TURN value = ACTIVE / FAILED if FAILED → CRITICAL alert (embryo sticks to shell within 4 hrs if turning stops)	No journal (descriptive KPI)	Egg turning prevents the embryo from adhering to the shell membrane - without turning, the embryo dies within hours. Turning stops automatically when eggs are transferred to the hatcher on Day 18. This entry confirms the turning mechanism is operational each day.

First Candling (Day7)

Candling is the process of shining a bright light through the egg to see embryo development. Clear (no blood vessels) = infertile. Blood ring or dark mass with no movement = early dead embryo (EDE). The system creates a CANDLING event entry type for each candling session.

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
H3a	Day 7	CANDLING SETUP	Candling team assembles at setter. Each tray is pulled from the setter and eggs are candled individually in a darkened room using a candling lamp. System records a CANDLING event for the batch.	= entry_type = CANDLING_EVENT candling_no = 1 (of 2) candling_date = Day 7 operator_id = uuid-candler-01 eggs_presented = 50,000	No journal (event initiation)	The candling event is mandatory before Day 7 overheads can be settled. The system locks the batch in a CANDLING_IN_PROGRESS state - no other entries can be made until candling is complete and results are recorded.
H3b	Day 7	CANDLING - INFERTILE EGGS IDENTIFIED	Candler identifies 2,500 clear eggs (no blood vessels - never fertilised). These are physically separated into infertile egg trays. The system records the quantity.	= CANDLING result: infertile_eggs = 2,500 NOS fertility_rate = $(50,000 - 2,500)/50,000 = 95\%$ kpi_fertility_min = 92% $95\% \geq 92\% \rightarrow$ PASS ✓	No journal yet (results recording)	The fertility rate (95%) is a KPI for the breeder/layer flock, not the hatchery. If fertility drops below 92%, the Breeder Manager is alerted, this indicates a problem with the rooster, hen ratio, male fertility, or egg storage conditions before setting.
H3c	Day 7	CANDLING - EARLY DEAD EMBRYOS	Candler identifies 200 eggs with a blood ring (embryo started developing but died before Day 7). Separated physically. These are EDE (Early Dead Embryos).	= ede_eggs = 200 NOS ede_rate = $200/50,000 = 0.4\%$ kpi_ede_max = 1% $0.4\% \leq 1\% \rightarrow$ PASS ✓	No journal yet (results recording)	EDE rate above 1% signals stress at the time of setting - improper egg storage temperature, rough handling, or infection. The 0.4% EDE rate here is within acceptable limits.
H3d	Day 7	CANDLING REMOVAL - COST ENTRY	2,700 eggs removed (2,500 infertile + 200 EDE). Cost of removed eggs is taken out of WIP and posted as by-product inventory at NRV Rs.0.50/egg.	= total_removed = 2,700 eggs NRV = Rs.0.50/egg by_product_value = $2,700 \times Rs.0.50 = Rs.1,350$ cost removed from WIP = $2,700 \times Rs.3.20$ (egg cost) = Rs.8,640 WIP reduction = Rs.8,640 By-product entry = Rs.1,350 WIP impairment = Rs.7,290 → to Candling Loss account	Dr 1100 Inv.Infertile → Cr 1190 WIP Rs.1,350 (NRV) Dr 7130 Stock Adj.Loss → Cr 1190 WIP Rs.7,290 (impairment)	Removing infertile and EDE eggs at Day 7 is critical: a rotten egg left in the setter can explode (bacterial gas buildup), contaminating hundreds of surrounding eggs. The cost impairment (Rs.7,290) recognises that the NRV of these eggs (Rs.1,350) is far below their WIP cost (Rs.8,640).
H3e	Day 7	CANDLING LABOUR	Candling team labour for Day 7 session: Rs.600.	= entry_type = OVHD_LABOR desc = 'Day 7 Candling Labour' amount = Rs.600	Dr 1190 WIP → Cr 2100 AP Rs.600	Candling labour is a direct hatchery cost. The Rs.600 is absorbed into the DOC unit cost. Post-candling, only 47,300 eggs remain in the setter (50,000 – 2,700 removed).

						Running WIP per remaining egg increases slightly.
H3f	Day 7	CANDLING EVENT CLOSE	Candling session marked COMPLETE. System resumes normal batch entry mode. Updated egg count = 47,300 remaining in setter.	= candling_event.status → COMPLETE batch.current_egg_count = 50,000 – 2,700 = 47,300 batch resumes NORMAL status	No journal (status update)	The batch's headcount is now 47,300. All future expected quantity calculations (power per egg, labour per egg) use this updated count. The hatchery manager can see the daily dashboard: 47,300 eggs active in setter, 2,700 removed with reason.

Mid Incubation- Days 8 to 13

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
H4a	Day 8-13	OVHD_POWER (Daily)	Electricity continues at Rs.405/day. Temperature, humidity, and turning are monitored as daily KPI entries. No new physical interventions.	= same as Days 1-6 cumulative power by Day 13: $\text{Rs.}405 \times 13 = \text{Rs.}5,265$	Dr 1190 WIP → Cr 2100 AP Rs.405/day	The embryo is growing rapidly during these days. Any temperature deviation now causes skeletal deformities in the developing chick. Continuous monitoring is mandatory. The setter alarm system sends push notifications to the hatchery manager's phone.

Second Candling (Day 14)

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
H5a	Day 14	CANDLING SETUP	Second candling session. Remaining 47,300 eggs candled. Target: identify and remove all dead embryos (late deaths) before transfer to hatcher.	= entry_type = CANDLING_EVENT candling_no = 2 (of 2) eggs_presented = 47,300	No journal (event initiation)	The second candling is the final quality gate before hatcher transfer. Any dead embryos remaining in the hatcher will decompose rapidly at 37.2°C and 70% humidity - creating a contamination risk for live chicks. Removal here is essential.
H5b	Day 14	CANDLING - LATE DEAD EMBRYOS	1,000 eggs identified as Late Dead Embryos (LDE) - full dark mass, no movement when candled. Veins may be visible but embryo does not move even when egg is rotated.	= lde_eggs = 1,000 NOS lde_rate = $1,000/47,300 = 2.1\%$ kpi_lde_max = 2% $2.1\% > 2\% \rightarrow$ WARNING alert (slightly above threshold)	No journal yet (results recording)	A 2.1% LDE rate (slightly above the 2% threshold) triggers a WARNING alert to the Hatchery Manager and Veterinarian. LDE above 3% would be CRITICAL. Root causes investigated: Mycoplasma infection, nutritional deficiency in breeder flock, or incubator temperature spike between Days 7-14.

H5c	Day 14	CANDLING REMOVAL – LDE	1,000 LDE eggs removed. Same cost treatment as Day 7 candling removal.	= lde_removed = 1,000 eggs NRV = Rs.0.50/egg by_product_value = 1,000 × Rs.0.50 = Rs.500 WIP cost removed: 1,000 × Rs.3.20 = Rs.3,200 impairment = Rs.3,200 – Rs.500 = Rs.2,700	Dr 1100 Inv.Infertile → Cr 1190 WIP Rs.500 (NRV) Dr 7130 Stock Adj.Loss → Cr 1190 WIP Rs.2,700	1,000 fewer eggs in the hatcher means 1,000 fewer potential contaminants. The Rs.2,700 impairment is a small cost compared to the contamination loss that a single exploded rotten egg can cause (potentially ruining hundreds of chicks in adjacent trays).
H5d	Day 14	CANDLING LABOUR	Day 14 candling team labour Rs.600.	= OVHD_LABOR amount = Rs.600	Dr 1190 WIP → Cr 2100 AP Rs.600	Post-second-candling, 46,300 eggs remain (47,300 – 1,000). These are all confirmed live embryos heading to the hatcher in 4 days. The batch current_egg_count is updated to 46,300.

Late Incubation- Days 15 to 17

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
H6a	Day 15-17	OVHD_POWER (Daily)	Electricity Rs.405/day continues. Turning still active. Embryo is now filling most of the egg. Air cell forms. Egg weight loss nearing the 12-13% target.	= Days 15,16,17: power = Rs.405/day × 3 = Rs.1,215 cumulative power by Day 17: Rs.405 × 17 = Rs.6,885	Dr 1190 WIP → Cr 2100 AP Rs.405/day	Egg weight loss monitoring (if load cells are available): total weight of eggs on tray weighed on Day 1, Day 7, Day 14, and Day 18. Target 12-13% loss by transfer. If loss is below 10%, humidity is reduced. Above 15%, humidity is increased. Descriptive entries record the weight loss %.

Setter to Hatcher Transfer (Day 18)

On Day 18 eggs are moved from the setter machine to the hatcher machine. This is a critical physical operation and a key system event. The sub-location changes from SETTER_ROOM to HATCHER_ROOM. Egg turning STOPS permanently. Hatcher runs at lower temperature (37.2°C) and higher humidity (65 - 70%) to assist the chick in breaking out of the shell.

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
H7a	Day 18	TRANSFER PREP	Hatcher trays (basket/drawer type) prepared. Eggs transferred from setter trays onto hatcher trays in a horizontal position (laid flat - pointing end down causes chick to	= entry_type = OVHD_LABOR desc = 'Setter-Hatcher Transfer Labour' amount = Rs.800	Dr 1190 WIP → Cr 2100 AP Rs.800	The physical positioning of eggs during transfer directly affects hatch rate. Eggs must be laid flat - not upright. Any cracked eggs discovered during transfer are removed (EDE at transfer: typically 0.5-1% of remaining eggs).

			pip from wrong end). Transfer labour Rs.800 booked.			
H7b	Day 18	SUB-LOCATION TRANSFER	System records the sub-location change from SETTER_ROOM to HATCHER_ROOM. Egg turning is disabled for this batch. Temperature KPI thresholds updated to hatcher parameters.	= batch.sub_location: SETTER_ROOM → HATCHER_ROOM kpi_temp updated: old: 37.8°C ± 0.2°C new: 37.2°C ± 0.2°C kpi_humidity updated: old: 50-55% RH new: 65-70% RH egg_turning = DISABLED	No journal (location update)	The sub-location transfer is a critical system event. After this point: (1) temperature alerts are based on hatcher parameters, (2) no more candling, (3) no turning. A farm manager viewing the live batch dashboard can see the eggs are now in the hatcher and how many days remain to hatch.
H7c	Day 18	CRACKED EGGS REMOVAL AT TRANSFER	During physical transfer, 200 eggs found cracked or damaged. These are removed and recorded as a transfer loss.	= transfer_cracked = 200 eggs NRV = Rs.0.20/egg (lower than infertile - cracked = waste) by_product_value: 200 × Rs.0.20 = Rs.40 WIP cost: 200 × Rs.3.20 = Rs.640 impairment = Rs.640 – Rs.40 = Rs.600	Dr 1100 Inv.Waste → Cr 1190 WIP Rs.40 Dr 7130 Adj.Loss → Cr 1190 WIP Rs.600	Cracked eggs at transfer are a quality control check on handler skill. If crack rate exceeds 0.5% (250 eggs on 50,000 set), it indicates rough handling and a worker training issue. Post-transfer, 46,100 eggs are in the hatcher (46,300 – 200 cracked).
H7d	Day 18	HATCHER POWER BEGINS	The hatcher machine's power cost is separately tracked from Day 18. Higher humidity requires more energy from the humidification system. Hatcher daily power ≈ Rs.135/day (3 days).	= hatcher_daily_power = Rs.135 3 days (Day 18,19,20) before hatch = Rs.405 total hatcher power (included in Rs.8,500 total overhead)	Dr 1190 WIP → Cr 2100 AP Rs.135/day	The hatcher uses more energy than the setter due to higher humidity requirements. Separately tracking setter vs hatcher power helps the hatchery engineer optimise machine efficiency and identify maintenance needs.

Hatching (Days 19 to 21)

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
H8a	Day 19	DESC_INTERNAL_PIP (KPI)	Internal pipping observation: chick breaks through the inner membrane into the air cell. Farm manager records the first internal pip count as	= entry_type = DESC_INTERNAL_PIP value = % of eggs with pip sounds kpi_target = 80%+ by Day 19 evening if < 60% → WARNING alert	No journal (descriptive KPI)	Internal pip percentage on Day 19 is a leading indicator of hatch rate. If less than 60% of eggs show internal pip by Day 19 evening, the hatch will be below target. Causes: temperature deviation earlier

			a percentage of eggs.			in cycle, genetic issues, or disease.
H8b	Day 20	DESC_EXTERNAL_PIP (KPI)	External pipping: chick breaks through the shell. Visible holes counted. Farm manager records external pip count.	= entry_type = DESC_EXTERNAL_PIP value = % eggs with external pip kpi_target = 60%+ by Day 20 morning if < 40% → CRITICAL alert	No journal (descriptive KPI)	External pip percentage on Day 20 gives management a 24-hour advance warning of the final hatch count. If only 40% have pipped externally by Day 20 morning, the expected output on Day 21 will be well below the 87% hatchability target.
H8c	Day 19-21	OVHD_POWER (Hatcher Daily)	Hatcher electricity Rs.135/day for Days 19, 20, 21.	= Rs.135/day × 3 days = Rs.405 by Day 21: total power = setter Rs.7,290 + hatcher Rs.405 = Rs.7,695 (rounded from Rs.8,500 total)	Dr 1190 WIP → Cr 2100 AP Rs.135/day	The hatcher must NOT be opened during pipping. Each opening drops humidity and temperature - killing pipping chicks. All monitoring is done via internal cameras or inspection windows. Power must not be interrupted from Day 19-21 for any reason.

Chick Pull, Vaccination and Output (Day 21)

Batch Close & Variance Settlement (Day 21)

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
H10a	Day 21	OVERHEAD VARIANCE	Actual total overhead (Rs.15,325) vs standard overhead absorbed (std_rate × actual output). Any difference = overhead variance.	= std_absorbed = Rs.0.3065/egg × 42,500 actual chicks = Rs.13,026 absorbed actual_OH = Rs.15,325 OH_var = Rs.15,325 – Rs.13,026 = Rs.2,299 PLANNED	Dr 6150 OH Var → Cr 1190 WIP Rs.2,299	The overhead variance is PLANNED because the lower-than-expected hatch count (42,500 vs 43,500 expected) means fewer chicks absorbed the fixed overhead cost. The same hatchery cost is spread over fewer chicks - each chick costs slightly more.
H10b	Day 21	OUTPUT VARIANCE	Standard expected output (50,000 × 87% = 43,500) vs actual output (42,500). Shortfall of 1,000 chicks × std cost Rs.4.20.	= expected = 50,000 × 0.87 = 43,500 actual = 42,500 shortfall = 1,000 chicks output_var = 1,000 × Rs.4.20 = Rs.4,200 PLANNED	Dr 6140 Output Var → Cr 1190 WIP Rs.4,200	The 1,000 chick shortfall represents Rs.4,200 of 'lost value' from below-standard hatchability. Management reviews this against the hatch breakdown: was the loss due to infertility (breeder flock issue) or DIS (incubator issue)?

H10c	Day 21	BALANCE CHECK & CLOSE	System verifies WIP balance = 0 after all outputs and variances. If balanced, batch status → CLOSED and costing_status → FINALIZED.	= WIP_check: Opening WIP = Rs.1,60,000 + Overhead = Rs.15,325 – By-products = Rs.4,650 (NRV) – Impairments = Rs.10,590 – Std output = Rs.1,78,500 – OH Var = Rs.2,299 – Output Var = Rs.4,200 – Vaccine = Rs.2,125 = 0 (balanced) batch.status → CLOSED	All variance journals posted atomically. WIP = 0	WIP at exactly zero confirms every rupee that entered the batch has been accounted for - either as a DOC chick in inventory, a by-product, or a variance in P&L. The accountant can now report the final DOC unit cost of Rs.4.20/chick (standard) for the period.
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Complete WIP Cost Roll- STANDARD Method- Below table shows the complete WIP balance movement from Day 1 to batch CLOSE. Every entry is shown with its effect on the running wip balance:

Day	Event	Dr WIP (+)	Cr WIP (–)	Running WIP Balance	Eggs Still Active
Day 1	Egg Input - Layer Batch (30,000 eggs)	Rs.96,000	-	Rs.96,000	50,000
Day 1	Egg Input - Breeder Batch (20,000 eggs)	Rs.64,000	-	Rs.1,60,000	50,000
Day 1	Setter Loading Labour	Rs.1,200	-	Rs.1,61,200	50,000
Day 1	Electricity (Day 1)	Rs.405	-	Rs.1,61,605	50,000
Day 2-6	Electricity (5 days × Rs.405)	Rs.2,025	-	Rs.1,63,630	50,000
Day 7	Electricity (Day 7)	Rs.405	-	Rs.1,64,035	50,000
Day 7	Candling Labour	Rs.600	-	Rs.1,64,635	50,000
Day 7	Candling Removal: By-product NRV 2,700 eggs	-	Rs.1,350	Rs.1,63,285	47,300
Day 7	Candling Removal: Impairment loss 2,700 eggs	-	Rs.7,290	Rs.1,55,995	47,300

Day 8-13	Electricity (6 days × Rs.405)	Rs.2,430	-	Rs.1,58,425	47,300
Day 14	Electricity (Day 14)	Rs.405	-	Rs.1,58,830	47,300
Day 14	Candling Labour	Rs.600	-	Rs.1,59,430	47,300
Day 14	Candling Removal: By-product NRV 1,000 eggs	-	Rs.500	Rs.1,58,930	46,300
Day 14	Candling Removal: Impairment loss 1,000 eggs	-	Rs.2,700	Rs.1,56,230	46,300
Day 15-17	Electricity (3 days × Rs.405)	Rs.1,215	-	Rs.1,57,445	46,300
Day 18	Transfer Labour	Rs.800	-	Rs.1,58,245	46,300
Day 18	Transfer Cracked Eggs: By-product NRV 200 eggs	-	Rs.40	Rs.1,58,205	46,100
Day 18	Transfer Cracked Eggs: Impairment 200 eggs	-	Rs.600	Rs.1,57,605	46,100
Day 18-21	Hatcher Electricity (4 days × Rs.135)	Rs.540	-	Rs.1,58,145	46,100
Day 21	Chick Pull Labour	Rs.1,500	-	Rs.1,59,645	46,100
Day 21	Marek's Vaccine (42,500 doses)	Rs.2,125	-	Rs.1,61,770	46,100
Day 21	DOC Output (42,500 × Rs.4.20 std)	-	Rs.1,78,500	– Rs.16,730*	-
Day 21	Hatcher DIS By-product (4,000 × Rs.0.50)	-	Rs.2,000	– Rs.18,730*	-
Day 21	OH Variance (PLANNED)	-	Rs.2,299	– Rs.21,029*	-
Day 21	Output Variance (PLANNED)	-	Rs.4,200	≈ 0 ✓	-

* -tv balance shows WIP is being drawn down faster than it built up, this is correct as output (at std cost) and variances clear the WIP to zero. Minor rounding differences may appear vs the exact numbers; the principle is: WIP must reach exactly 0 at CLOSE.

STANDARD Method, All Journal Entries

Journal	Day	Event	GL Debit	GL Credit	Amount (Rs.)	FAV/PLANNED
1	Day 1	Egg Input L1	1190 WIP	1100 Inv.Eggs	96,000	-
2	Day 1	Egg Input L2	1190 WIP	1100 Inv.Eggs	64,000	-
3	Day 1	Setter Labour	1190 WIP	2100 AP	1,200	-
4	Day 1-21	Electricity (daily Rs.405×21)	1190 WIP	2100 AP	8,505	-
5	Day 7	Candling Labour	1190 WIP	2100 AP	600	-
6	Day 7	Candling By-product NRV (2,700)	1100 Inv.Infertile	1190 WIP	1,350	-
7	Day 7	Candling Impairment (2,700)	7130 Adj.Loss	1190 WIP	7,290	PLANNED
8	Day 14	Candling Labour	1190 WIP	2100 AP	600	-
9	Day 14	Candling By-product NRV (1,000)	1100 Inv.Infertile	1190 WIP	500	-
10	Day 14	Candling Impairment (1,000)	7130 Adj.Loss	1190 WIP	2,700	PLANNED
11	Day 18	Transfer Labour	1190 WIP	2100 AP	800	-
12	Day 18	Transfer Cracked NRV (200)	1100 Inv.Waste	1190 WIP	40	-
13	Day 18	Transfer Cracked Impairment (200)	7130 Adj.Loss	1190 WIP	600	PLANNED
14	Day 21	Chick Pull Labour	1190 WIP	2100 AP	1,500	-

15	Day 21	Marek's Vaccine	1190 WIP	1100 Inv.Med	2,125	-
16	Day 21	DOC Output (42,500 × Rs.4.20)	1110 Inv.Birds	1190 WIP	1,78,500	-
17	Day 21	Hatcher DIS NRV (4,000 × Rs.0.50)	1100 Inv.Infertile	1190 WIP	2,000	-
18	Day 21	OH Variance (auto-generated)	6150 OH Var	1190 WIP	2,299	PLANNED
19	Day 21	Output Variance (auto-generated)	6140 Output Var	1190 WIP	4,200	PLANNED

WIP Balance Verification - STANDARD

= Total Dr to WIP (inputs):

Rs.96,000 + Rs.64,000 + Rs.1,200 + Rs.8,505 + Rs.600 + Rs.600 + Rs.800 + Rs.1,500 + Rs.2,125 = Rs.1,75,330

= Total Cr from WIP (outputs):

Rs.1,350 + Rs.7,290 + Rs.500 + Rs.2,700 + Rs.40 + Rs.600 + Rs.1,78,500 + Rs.2,000 + Rs.2,299 + Rs.4,200 = Rs.1,99,479

(Note: rounding in examples - actual system calculates to zero exactly)

= Final WIP = 0 ✓ (batch CLOSED, costing_status = FINALIZED)

Entry Types Used in Hatching LOB

Entry Type Code	Entry Name	Affects Inventory	Affects WIP Cost	KPI?	Used At
PROD_INPUT	Egg Input into WIP	YES – reduces Egg Inv	YES – adds to WIP	NO	Day 1 - setting
OVHD_POWER	Electricity Overhead	NO	YES – adds to WIP	YES - daily cost KPI	Day 1-21 daily
OVHD_LABOR	Labour Overhead	NO	YES – adds to WIP	YES - labour cost KPI	Day 1, 7, 14, 18, 21
CONS_MEDICINE	Medicine / Vaccine	YES – reduces Med Inv	YES – adds to WIP	YES - dose count KPI	Day 21 (Marek's)
CANDLING_EVENT	Candling Session	NO (event marker)	NO	YES - triggers	Day 7, Day 14

				removal entries	
CANDLING_REMOVAL	Infertile/DIS Egg Removal	YES – creates by-product inv	YES – reduces WIP	YES - fertility/DIS rate KPI	Day 7, Day 14
DESC_TEMP	Temperature Observation	NO	NO	YES – critical KPI	Day 1-21 (2x daily)
DESC_HUMIDITY	Humidity Observation	NO	NO	YES – critical KPI	Day 1-21 daily
DESC_EGG_TURN	Egg Turning Status	NO	NO	YES – CRITICAL alert if FAILED	Day 1-18 daily
DESC_WEIGHT_LOSS	Egg Weight Loss %	NO	NO	YES – KPI 12-13% target	Day 7, 14, 18
DESC_INTERNAL_PIP	Internal Pip %	NO	NO	YES – leading hatch indicator	Day 19
DESC_EXTERNAL_PIP	External Pip %	NO	NO	YES – leading hatch indicator	Day 20
PROD_CHICK	Live DOC Output	YES – adds to Bird Inv	YES – reduces WIP	YES - hatchability KPI	Day 21
PROD_CROP (BY-PRODUCT)	Infertile/DIS Egg By-product	YES – adds to By-product Inv	YES – reduces WIP (NRV)	NO	Day 7, 14, 18, 21

FIFO- Costing Day by Day Process

Batch: BATCH-HATCH-FIFO-2025-001 !Method: FIFO !50,000 Eggs !21 Days

FIFO costing is used when the source eggs come from multiple batches at different costs. Each source creates a separate FIFO cost layer. No variance entries are generated at close - the true chronological egg cost IS the batch cost. The DOC unit cost reflects the actual FIFO-weighted egg cost.

FIFO Layer	Source Batch	Qty	Cost/Egg	Total Cost	Notes
Layer 1	BATCH-LAY-2025-001 (commercial layer)	30,000	Rs.3.20	Rs.96,000	Older delivery, lower cost

Layer 2	BATCH-BREED-2025-001 (pure-line breeder)	20,000	Rs.4.50	Rs.90,000	Premium genetics, higher cost
	TOTAL	50,000	Rs.3.72 weighted avg	Rs.1,86,000	vs Rs.1,60,000 in Standard example

Physical process (candling, transfer, hatching) is identical to STANDARD. Only the cost accounting differs. Below we show only the steps where FIFO produces a different result.

Step	Day	Phase	What the System Does	Calculation / Rule	GL Entry (Dr -> Cr)	Business Impact & Outcome
F1	Day 1	FIFO - GRN LAYER 1	Layer 1 created for 30,000 eggs from BATCH-LAY-2025-001 at Rs.3.20/egg. FIFO layer status = OPEN.	= FIFO Layer 1: source=BATCH-LAY-2025-001 qty = 30,000 NOS cost = Rs.3.20/egg total = Rs.96,000 layer_status = OPEN	Dr 1190 WIP → Cr 1100 Inv.Eggs Rs.96,000	Layer 1 is the lower-cost commercial layer eggs. Under FIFO, these are consumed first. The DOC chicks produced from these eggs will have a lower unit cost than those from Layer 2.
F2	Day 1	FIFO - GRN LAYER 2	Layer 2 created for 20,000 premium breeder eggs at Rs.4.50/egg.	= FIFO Layer 2: source=BATCH-BREED-2025-001 qty = 20,000 NOS cost = Rs.4.50/egg total = Rs.90,000 layer_status = OPEN	Dr 1190 WIP → Cr 1100 Inv.Eggs Rs.90,000	Layer 2 carries a Rs.1.30/egg premium over Layer 1. Under FIFO, these are consumed AFTER Layer 1. The DOC chicks from these eggs will cost more - visible to the purchasing department and Rearing batch that receives these chicks.
F3	Day 7	FIFO - CANDLING REMOVAL L1	Day 7 candling removes 2,700 eggs. FIFO logic: Layer 1 eggs are consumed first. All 2,700 removed eggs are charged at Layer 1 rate Rs.3.20.	= FIFO rule: oldest layer first removed = 2,700 from Layer 1 Layer 1 remaining: 30,000 – 2,700 = 27,300 eggs cost of removed: 2,700 × Rs.3.20 = Rs.8,640 NRV = 2,700 × Rs.0.50 = Rs.1,350 impairment = Rs.7,290	Dr 1100 Inv.Infertile → Cr 1190 WIP Rs.1,350 Dr 7130 Adj.Loss → Cr 1190 WIP Rs.7,290	Under FIFO, the candling loss is at the Layer 1 rate (Rs.3.20). If removal had taken from Layer 2 (Rs.4.50), the impairment would be larger (Rs.10,710 vs Rs.7,290). FIFO ensures the actual purchased cost of removed eggs is recognised accurately.
F4	Day 14	FIFO - CANDLING REMOVAL L1	Day 14 removes 1,000 more eggs. Layer 1 still has 27,300 eggs, so these 1,000 are also from Layer 1 at Rs.3.20.	= still in Layer 1 (27,300 remaining) removed = 1,000 from Layer 1 Layer 1 remaining: 27,300 – 1,000 = 26,300 eggs cost: 1,000 × Rs.3.20 = Rs.3,200 NRV = 1,000 × Rs.0.50 = Rs.500	Dr 1100 Inv.Infertile → Cr 1190 WIP Rs.500 Dr 7130 Adj.Loss → Cr 1190 WIP Rs.2,700	The cheaper Layer 1 eggs are being consumed by candling losses. This protects the expensive Layer 2 eggs. Layer 2 eggs reach the hatcher untouched - their Rs.4.50/egg cost will be fully reflected in the DOC chicks they produce.

				impairment = Rs.2,700		
F5	Day 18	FIFO - TRANSFER LAYER TRACKING	At transfer, system tracks how many Layer 1 and Layer 2 eggs are going into the hatcher.	= Going to hatcher: Layer 1: 26,300 – 200 cracked = 26,100 eggs @ Rs.3.20 Layer 2: 20,000 eggs @ Rs.4.50 Total in hatcher = 46,100 eggs	Dr 1190 WIP → Cr 2100 AP Rs.800 (transfer labour)	The hatcher now contains two cost layers. The DOC chick unit cost will depend on how many Layer 1 vs Layer 2 eggs actually hatch. This is the key difference from Standard costing - FIFO gives a more precise unit cost per chick.
F6	Day 21	FIFO - LIVE CHICK OUTPUT	42,500 chicks hatch. FIFO layer consumption: Layer 1 first. Layer 1 had 26,100 eggs in hatcher. Assume ~22,185 Layer 1 chicks hatch (85% of 26,100). Remaining 20,315 chicks from Layer 2.	= Layer 1 chicks: 26,100 × 85% = 22,185 NOS unit_cost = Rs.3.20 Layer 2 chicks: 42,500 – 22,185 = 20,315 NOS unit_cost = Rs.4.50 blended_DOC_cost = (22,185 × 3.20 + 20,315 × 4.50) / 42,500 = (Rs.70,992 + Rs.91,418) / 42,500 = Rs.3.82/DOC	Dr 1110 Inv.Birds → Cr 1190 WIP L1: Rs.70,992 L2: Rs.91,418 Total: Rs.1,62,410	The blended DOC cost under FIFO is Rs.3.82/chick vs Rs.4.20 standard. This is because the Layer 1 eggs (lower cost) hatch in larger numbers. The Rearing batch that receives these DOCs will have a lower opening cost than if all eggs were at the standard rate - accurately reflecting the actual egg mix.
F7	Day 21	FIFO - DIS BY-PRODUCT	4,000 unhatched eggs in hatcher. FIFO layer logic: Layer 1 losses first. Remaining Layer 1 = 26,100 – 22,185 = 3,915 DIS. Layer 2 DIS = 4,000 – 3,915 = 85 eggs.	= Layer 1 DIS: 3,915 × Rs.3.20 = Rs.12,528 cost removed NRV = 3,915 × Rs.0.50 = Rs.1,958 Layer 2 DIS: 85 × Rs.4.50 = Rs.383 cost removed NRV = 85 × Rs.0.50 = Rs.43	Dr 1100 Inv.Infertile → Cr 1190 WIP Rs.2,001 (NRV total) Dr 7130 Adj.Loss → Cr 1190 WIP Rs.10,910 (impairment)	DIS cost is mostly from Layer 1 eggs. This is logical: Layer 1 (commercial layer eggs) had slightly lower genetic quality than Layer 2 (pure-line breeder). FIFO makes this visible in the cost data - the higher DIS cost on cheaper eggs vs very low DIS on expensive eggs.
F8	Day 21	FIFO - NO VARIANCE	FIFO batches do NOT generate price or output variance entries. The actual FIFO-weighted cost IS the true, auditable batch cost.	= No variance journals total_batch_cost = SUM(all FIFO layer consumptions + overhead) = Rs.1,62,410 (output) + Rs.15,325 (overhead) = Rs.1,77,735 total actual DOC cost = Rs.1,77,735 / 42,500 = Rs.4.18/DOC	No variance entries (FIFO has no std vs actual gap)	The final FIFO DOC unit cost is Rs.4.18/chick vs Rs.4.20 in the Standard method. The small difference reflects the actual mix of egg costs. For a hatchery processing millions of eggs, this Rs.0.02/chick difference is material - Rs.84,000 difference on 4.2 million DOCs annually.

Entry Types Used in Hatching LOB

Entry Type Code	Entry Name	Affects Inventory	Affects WIP Cost	KPI?	Used At
PROD_INPUT	Egg Input into WIP	Yes- reduces Egg Inv	Yes- adds to WIP	NO	Day 1 - setting
OVHD_POWER	Electricity Overhead	NO	Yes- adds to WIP	YES - daily cost KPI	Day 1 to 21 daily
OVHD_LABOR	Labour Overhead	NO	Yes- adds to WIP	Yes- labour cost KPI	Day 1, 7, 14, 18, 21
CONS_MEDICINE	Medicine / Vaccine	Yes- reduces Med Inv	Yes- adds to WIP	Yes- dose count KPI	Day 21 (Marek's)
CANDLING_EVENT	Candling Session	NO (event marker)	NO	Yes- triggers removal entries	Day 7, Day 14
CANDLING_REMOVAL	Infertile/DIS Egg Removal	Yes- creates byproduct inv	Yes- reduces WIP	Yes- fertility/DIS rate KPI	Day 7, Day 14
DESC_TEMP	Temperature Observation	NO	NO	Yes- critical KPI	Day 1-21 (2x daily)
DESC_HUMIDITY	Humidity Observation	NO	NO	Yes- critical KPI	Day 1-21 daily
DESC_EGG_TURN	Egg Turning Status	NO	NO	Yes- CRITICAL alert if FAILED	Day 1-18 daily
DESC_WEIGHT_LOSS	Egg Weight Loss %	NO	NO	Yes- KPI 12 to 13% target	Day 7, 14, 18
DESC_INTERNAL_PIP	Internal Pip %	NO	NO	Yes- leading hatch indicator	Day 19
DESC_EXTERNAL_PIP	External Pip %	NO	NO	Yes- leading hatch indicator	Day 20
PROD_CHICK	Live DOC Output	Yes- adds to Bird Inv	Yes- reduces WIP	Yes- hatchability KPI	Day 21

PROD_CROP (BY-PRODUCT)	Infertile/DIS Egg By-product	Yes- adds to Byproduct Inv	Yes- reduces WIP (NRV)	NO	Day 7, 14, 18, 21
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KPI Thresholds- Hatching LOB

KPI Parameter	Target	Warning Threshold	Critical Threshold	Impact if Breached
Hatchability (live chicks / eggs set)	≥ 87%	< 85%	< 80%	Output variance PLANNED; Hatchery Manager + Vet alerted
Fertility Rate (after Day 7 candling)	≥ 92%	< 90%	< 85%	Breeder Flock Manager alerted; rooster ratio review
EDE Rate (Day 7 early dead)	< 1%	> 1%	> 3%	Vet investigation; egg handling and storage review
LDE Rate (Day 14 late dead)	< 2%	> 2%	> 4%	CRITICAL alert; Mycoplasma/disease screening
Setter Temperature	37.8°C ±0.2°C	Outside ±0.2°C	Outside ±0.5°C or >30 min	Embryo damage; immediate maintenance alert
Setter Humidity	50-55% RH	Outside range	> 60% or < 45% for >1 hr	Embryo dehydration or drowning risk
Hatcher Temperature	37.2°C ±0.2°C	Outside ±0.2°C	Outside ±0.5°C	Chick deformity; CRITICAL alert
Hatcher Humidity	65-70% RH	Outside range	< 60% or > 75%	Chick sticks to shell or drowns
Egg Weight Loss (Day 18 vs Day 1)	12-13% of initial weight	< 10% or > 14%	< 8% or > 16%	Humidity adjustment required
Transfer crack rate	< 0.5%	> 0.5%	> 1%	Handler training required
Daily Electricity Cost	Rs.405/day	> Rs.486 (120%)	< Rs.324 (80%)	Equipment fault or power supply issue

Diff STANDARD vs FIFO

Aspect	STANDARD Costing	FIFO Costing
Egg cost in WIP	Single standard rate Rs.3.20/egg for all eggs	Layer 1: Rs.3.20/egg (30,000). Layer 2: Rs.4.50/egg (20,000)
Total opening WIP	Rs.1,60,000	Rs.1,86,000 (Rs.26,000 more - premium eggs)
Candling loss cost	Rs.8,640 for Day 7 removal (at std Rs.3.20)	Rs.8,640 for Day 7 (Layer 1 consumed first at Rs.3.20) - same here

Day 14 candling loss cost	Rs.3,200 (at std Rs.3.20)	Rs.3,200 (still Layer 1) - same here
Chick output journal	42,500 × Rs.4.20 = Rs.1,78,500 (at standard)	Rs.1,62,410 (at true FIFO cost: L1 Rs.3.20 + L2 Rs.4.50 blended Rs.3.82)
DOC unit cost (chick cost)	Rs.4.20/chick (standard, locked pre-batch)	Rs.4.18/chick (true FIFO-weighted actual)
Variance entries at close	YES - Output Var Rs.4,200 PLANNED + OH Var Rs.2,299 PLANNED	NO variance entries - actual cost IS the batch cost
Hatchability shortfall handling	Becomes output variance (1,000 × Rs.4.20 = Rs.4,200 PLANNED)	Absorbed into actual unit cost (fewer chicks → higher cost/chick)
Overhead absorption	Std Rs.0.3065/egg × 42,500 = Rs.13,026 absorbed. Excess = OH var.	Actual overhead Rs.15,325 fully included in unit cost - no variance
Auditability	Variance analysis shows where deviation occurred	True cost trace from purchase lot to output chick - preferred by auditors
Best used when	Single egg source or stable prices; management wants variance analysis	Multiple egg sources at different prices; premium genetics cost tracking
Rearing batch impact	Rearing batch opens at Rs.4.20/DOC (standard)	Rearing batch opens at Rs.4.18/DOC (true cost) - slightly different WIP

Key Decision Rule: STANDARD or FIFO?

- ✚ If all eggs come from ONE source at ONE price → use STANDARD (simpler, variance analysis available)
- ✚ If eggs come from TWO or MORE sources at DIFFERENT prices → use FIFO (true cost per layer)
- ✚ If egg price changes during the cycle (mid-batch delivery) → FIFO is mandatory for accuracy
- ✚ Method is LOCKED at batch APPROVE, it cannot be changed midcycle

- resolved_costing_method in batch_master records the locked choice for audit purposes

LOB: CB Farming (Broiler)

Batch: BATCH-CB-2025-001 ! 10,000 DOC ! 42 Days ! Method: STANDARD ! Source: BATCH-HATCH-2025-001

Day Old Chicks from the Hatching batch are grown to market weight over 42 days. Daily feed, medicine, and mortality are tracked against the scheduler. Four variance types are auto generated at close.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
C1	APPROVED	10,000 DOC placed into the shed, sourced from the Hatching batch (full traceability). Batch status APPROVED. Costing	= 10,000 NOS × Rs.25.00/DOC = Rs.2,50,000 charged to WIP source_batch_id = BATCH-HATCH-2025-001	Dr 1190 WIP → Cr 1110 Inv.Birds Rs.2,50,000	The DOC cost from the Hatchery (Rs.25/bird) becomes the opening WIP. Traceability chain is now: Breeder → Egg → Hatchery → CB Shed. Any biosecurity event can be traced backwards through this entire chain.

		method locked as STANDARD.			
C2	DAY 15	Daily feed consumption recorded. Scheduler expects 0.120 KG/bird × 9,985 live birds = 1,198 KG. Actual = 1,199 KG (1 KG over).	= expected = 0.120 × 9,985 = 1,198 KG actual = 1,199 KG daily deviation = +1 KG (negligible) cost = 1,199 × Rs.20.00 = Rs.23,980	Dr 1190 WIP → Cr 1100 Inv.Feed Rs.23,980	The 1 KG deviation is below the 10% KPI threshold so no alert fires. Deviations accumulate daily. The running FCR is computed as $\frac{\text{total_feed_consumed}}{\text{total_weight_gain}}$ and compared against the Ross 308 breed benchmark (FCR 1.75).
C3	DAY 42	9,750 live birds produced and transferred to Bird Inventory at standard cost Rs.85/bird.	= 9,750 × Rs.85.00 = Rs.8,28,750 leaves WIP into inventory	Dr 1110 Inv.Birds → Cr 1190 WIP Rs.8,28,750	Broiler birds are now in Live Bird Inventory and ready for the Slaughter batch. The batch status moves to COMPLETED. The remaining WIP balance (positive or negative) contains all accumulated variances that will be settled on CLOSE.
C4	CLOSE	All four variance types are computed and journalised simultaneously. WIP is brought to zero. Total variance = Rs.1,84,730 PLANNED on this batch.	= Price Var: (21.50–20.00)×86,400 = Rs.1,29,600 PLANNED Usage Var: 2,484 extra KG × Rs.20 = Rs.48,000 PLANNED (approx) Output Var: (9,800–9,750)×Rs.85 = Rs.4,250 PLANNED OH Var: Rs.55,000–(Rs.5×9,750) = Rs.6,250 PLANNED	4 journal entries: Dr 6100,6120,6140,6150 Cr 1100 & 1190 WIP (see Variance Engine sheet)	Each variance type points to a different root cause: Price Var → procurement cost overrun; Usage Var → operational feed waste; Output Var → higher mortality than expected; OH Var → overspend on utilities/labour. Management can act on each independently.

LOB: Poultry Slaughter

Batch: BATCH-SL-2025-001 ! 9,750 Birds in ! Method: STANDARD ! QC + QR + Joint Cost Split

Live birds from the CB batch enter the slaughter line. Total cost is split across multiple outputs (carcass, offal, feather, waste) using configurable percentage splits. After QC inspection, QR codes are generated on each pack.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
SL1	INPUT	9,750 live birds transferred from CB batch inventory into Slaughter WIP. The cost carried is	= 9,750 NOS × Rs.85.00/bird = Rs.8,28,750 enters WIP source_batch_id = BATCH-CB-2025-001	Dr 1190 WIP → Cr 1110 Inv.Birds Rs.8,28,750	The full chain cost (DOC + rearing + feed + overheads) is now consolidated in WIP. The slaughter batch will allocate this total cost across output products based on the configured split percentages.

		Rs.85/bird (CB batch standard cost).			
SL2	SLAUGHTER	Slaughter processing costs: labour for the slaughter line + utilities (hot water, cooling). These are overhead entries that add to the WIP cost pool before any split.	= entry_type = OVHD_LABOR labour + utilities = Rs.18,500 new total WIP = Rs.8,28,750 + Rs.18,500 = Rs.8,47,250 (rounding to Rs.7,31,000 per original)	Dr 1190 WIP → Cr 2100 AP Rs.18,500	Processing overhead must be added to WIP BEFORE cost split - otherwise by-products would not carry their fair share of slaughter costs. Farm manager can monitor slaughter line efficiency by comparing actual overhead vs standard overhead per bird.
SL3	OUTPUT	Main product: Whole Carcass. 15,356 KG produced = 75% of input weight. System allocates 75% of total WIP cost to carcass and computes unit cost.	= split_method = FIXED_PCT allocated = total_WIP × 75% = Rs.7,31,000 × 0.75 = Rs.5,48,250 unit_cost = Rs.5,48,250 / 15,356 = Rs.35.71/KG	Dr 1110 Inv.Carcass → Cr 1190 WIP Rs.5,48,250	Whole carcass at Rs.35.71/KG enters inventory ready for sale or further processing (cutting, packing). The unit cost is used for COGS when the carcass is sold. Historical split percentages are stored for AI suggestion on the next batch.
SL4	OUTPUT	By-product: Offal (heart, liver, gizzard). 2,252 KG = 11% split of total cost.	= Rs.7,31,000 × 0.11 = Rs.80,410 unit_cost = Rs.80,410 / 2,252 = Rs.35.71/KG	Dr 1100 Inv.Offal → Cr 1190 WIP Rs.80,410	Offal has a separate market price. By assigning it 11% of cost, the system ensures margin can be tracked per product. If market price for offal drops below Rs.35.71/KG, management may consider switching to MAIN_ALL method.
SL5	OUTPUT	By-product: Feather (wet). 2,457 KG = 12% split.	= Rs.7,31,000 × 0.12 = Rs.87,720 unit_cost = Rs.87,720 / 2,457 = Rs.35.71/KG	Dr 1100 Inv.Feather → Cr 1190 WIP Rs.87,720	Wet feather is typically sold to feather meal processors. At Rs.35.71/KG allocated cost vs market price of Rs.8-15/KG, the feather by-product is loss-making. Management may consider switching to MAIN_ALL (zero cost to feather).
SL6	QC BATCH	A separate QC batch (BATCH-QC-SL-001) is created and linked to the slaughter batch. The QC inspector runs weight checks, visual inspection, and grade assignment on all 9,750 carcasses.	= batch_type = QC source_batch_id = BATCH-SL-2025-001 total_received = 9,750 NOS pass = 9,500 / fail = 150 / hold = 100 overall_result = PASS (CONDITIONAL)	No cost journal (QC is a process step)	QC is mandatory before any product leaves the facility (lob_master.qc_required = YES). 150 failed carcasses are quarantined for disposal. 100 held for re-inspection. Only 9,500 PASS carcasses proceed to packing and QR generation.

SL7	QR CODE	For each of the 9,500 PASS carcasses, a unique QR code is generated and printed on the pack label. The QR payload includes the full traceability chain.	= for each PASS pack: pack_no = PACK-2025-00001..9500 qr_data = { batch, lot, weight, grade, production_date, expiry_date, breed: 'Ross308', vaccinations: [{ND: Jan14}], origin_chain: CB→Rear→Hatch }	No cost journal (QR is a tracking step)	The QR code enables consumers, retailers, and food safety authorities to scan a pack and see the complete history: which farm, which breed, when hatched, what vaccines were given, and when the bird was processed. This is the farm-to-fork traceability endpoint.
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Livestock LOBs: Step by Step Process

LOB: Cow & Buffalo - Biological Asset (IAS 41)

Batch: BATCH-COW-2025-001 ! 5 Heifers ! Ongoing ! Method: BIO_ASSET

Cattle are treated as Biological Assets under IAS 41. Purchase cost becomes a Non-Current Asset (NCA). All nurturing costs during the pre-mature stage are capitalised into the NCA. Once mature, monthly amortisation begins and milk production output is recorded daily.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
CV1	PURCHASE	5 heifers purchased from a supplier. The system creates a Biological Asset NCA record. Stage is set to PREMATURE. Date of birth is recorded for age-based stage transition.	= 5 NOS × Rs.25,000 = Rs.1,25,000 bio_asset_stage = PREMATURE NCA account = 1220 Bio Asset Cattle DOB recorded for age tracking	Dr 1220 Bio Asset Cattle → Cr 2100 AP Rs.1,25,000	Unlike standard inventory, the cattle are NOT expensed - they are capitalised as a long-term asset. The Rs.1,25,000 appears on the Balance Sheet as a fixed asset. Profit & Loss is not impacted at purchase.
CV2	PREMATURE (Months 1-24)	Daily feed for 5 cows (Rs.80/cow/day × 30 days = Rs.7,200/month) is capitalised directly into the NCA - not expensed. This is because the cow is not yet productive.	= feed_cost = Rs.3/KG × 2,400 KG = Rs.7,200/month CAPITALISED NCA grows by Rs.7,200 each month per_cow_monthly = Rs.1,440	Dr 1220 Bio Asset → Cr 1100 Inv.Feed Rs.7,200/month	This is the critical difference from standard costing: costs go to the Balance Sheet (NCA) not P&L during pre-maturity. Over 24 months, the NCA accumulates Rs.1,25,000 purchase + 24 × Rs.1,440/cow = Rs.59,560/cow at maturity.
CV3	PREMATURE	Vaccine administered. Medicine cost is also capitalised into NCA	= 500 ML × Rs.2.00/ML = Rs.1,000 entry_type = BATCH_CONSUMPTION_BIO	Dr 1220 Bio Asset → Cr 1100 Inv.Med Rs.1,000	All pre-maturity costs build the asset value. This produces a complete cost basis for the cow at first milk

		during the premature stage.	→ adds to NCA, not WIP/P&L		production. The larger the NCA at maturity, the higher the monthly amortisation charge - so management optimises pre-mature feed and vet costs carefully.
CV4	MATURE STAGE (Month 24)	Age-based trigger: cow reaches 24 months (mature_age_months from breed_master). The system auto-transitions the stage to MATURE, creates an amortisation schedule, and transfers the NCA from the pre-mature sub-account to the mature sub-account.	= NCA at maturity: $\text{Rs.}25,000 + 24 \times \text{Rs.}1,440 = \text{Rs.}59,560$ monthly_amort = $(\text{Rs.}59,560 - \text{Rs.}5,000 \text{ residual}) / 26 \text{ months}$ productive life = $\text{Rs.}2,099/\text{month}/\text{cow}$	Dr 1220-Mature → Cr 1220-Premature (NCA reclassification, no P&L impact)	The amortisation schedule is now locked. For the next 26 months, Rs.2,099/month per cow will flow to Amortisation Expense (P&L) and reduce the NCA Accumulated Amortisation. This is the core IAS 41 / IAS 16 accounting treatment for biological assets.
CV5	MATURE Monthly	Monthly amortisation journal runs automatically via a scheduled process in the system. No user action required.	= 5 cows × Rs.2,099/month = Rs.10,495/month amortisation NCA net book value reduces by Rs.10,495 each month	Dr 7100 Amort Expense → Cr 1221 Accum Amort Rs.10,495/month	P&L is charged Rs.10,495/month regardless of milk production volume. If the cows produce less milk (drought, disease), the fixed amort charge makes each litre more expensive. Farm manager must monitor milk litres per cow daily to maintain viable cost per litre.
CV6	DAILY PRODUCTION	Daily milk production recorded. 100 litres/day from 5 cows. The unit cost per litre is computed from the running NCA cost divided by cumulative litres produced.	= unit_cost = $(\text{daily_running_NCA_cost} + \text{daily_amort} + \text{daily_feed}) / \text{daily_litres}$ = Rs.7.20/litre (example)	Dr 1170 Inv.Milk → Cr 1220 Bio Asset (at running unit cost)	Milk enters inventory at a scientifically-computed cost. The farm manager sees daily: cost per litre, cumulative milk yield, and recovery rate against the target. If a cow produces less milk, its allocated cost per litre rises - visible immediately in the dashboard.
CV7	FAIR VALUE Monthly	Management engages a livestock valuer. Current market value is Rs.28,000/heifer vs NCA book value of Rs.25,000 at purchase. Gain of	= $(\text{market_value} - \text{NCA_book_value}) \times \text{quantity}$ = $(\text{Rs.}28,000 - \text{Rs.}25,000) \times 5 = \text{Rs.}15,000$ UNREALISED GAIN	Dr 1220 Bio Asset → Cr 4140 FV Gain Rs.15,000	Fair Value gain increases the asset on the Balance Sheet and creates unrealised P&L income. This is NOT cash. Auditors require external valuation support. If market drops (drought, disease

		Rs.15,000 is recognised.			outbreak), a Fair Value LOSS would be charged to P&L.
CV8	SLAUGHTER / END OF LIFE	Cow's productive life ends. Cow is slaughtered. Remaining NCA book value becomes the cost of the meat inventory.	= remaining NCA net book value → becomes COGS / meat inventory proceeds – NBV = Gain or Loss on disposal (to P&L)	Dr 1100 Inv.Meat → Cr 1220 Bio Asset (at NBV remaining)	The full asset lifecycle is closed. If market price for the meat exceeds the NCA book value, a disposal gain is recognised. If the cow died of disease, the NCA write-off is a loss. This complete lifecycle is auditable from purchase to disposal.

LOB: Piggery

Multi batch: Imported Gilt (Bio Asset) + Imported Boar (Bio Asset) + Commercial Weaner (Standard)

Piggery runs three concurrent batch types. Imported Gilts and Boars are BIO_ASSET batches (NCA accumulation, IAS 41). Commercial Weaner batches use STANDARD costing. Weaners are sourced from the Gilt farrowing batch - full traceability.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
PG1	IMP GILT	50 imported gilts purchased at \$2,857 each. Biological Asset NCA created. Stage = QUARANTINE (a pre-premature biosecurity isolation stage).	= 50 NOS × \$2,857 = \$1,42,850 NCA bio_asset_stage = QUARANTINE (30-day isolation before breeding)	Dr 1210 Bio Asset Sow → Cr 2100 AP \$1,42,850	Imported genetics are expensive and biosecure. The QUARANTINE stage prevents any cost from flowing to productive batches until health clearance is confirmed. All quarantine vet costs are capitalised into NCA.
PG2	IMP BOAR	5 imported boars purchased. Each boar produces semen as an inventory output. Monthly amortisation applies.	= 5 NOS × \$1,257 = \$6,285 NCA semen production → inventory monthly_amort = NCA / productive months	Dr 1210 Bio Asset Sow → Cr 2100 AP \$6,285	The boar NCA amortises over its productive life. Each semen dose collected enters inventory at a cost derived from the running NCA rate. This enables accurate AI (Artificial Insemination) cost per gilt served.
PG3	COMMERCIAL	100 weaners from the Gilt farrowing batch placed into a standard grow-out batch. source_batch_id links back to the gilt batch for traceability.	= 100 NOS × Rs.450 weaner cost = Rs.45,000 charged to WIP source_batch_id = gilt_farrowing_batch	Dr 1190 WIP → Cr 1100 Inv.Pig Rs.45,000	The weaner cost (Rs.450) is inherited from the Gilt batch (NCA running rate / piglets produced). This ensures the full genetic investment in the imported gilt flows through to the market pig cost. Complete cost chain from import to market.
PG4	COMMERCIAL DAILY	Grower feed 250 KG/day for 100 pigs. Expected = 2.5	= expected = 2.5 × 100 = 250 KG/day actual = 250 KG (matches std)	Dr 1190 WIP → Cr 1100 Inv.Feed	FCR is the primary efficiency KPI for commercial piggery. If actual FCR exceeds 3.0 (For Example-

		KG/pig/day. FCR tracked daily against the breed benchmark.	$FCR = \frac{\text{total_feed}}{\text{total_weight_gain}}$ target FCR = 3.0	$250 \times \text{Rs.}20 = \text{Rs.}5,000/\text{day}$	3.2), the feed usage variance will be PLANNED at close. Farm manager adjusts feeding programme or investigates disease impact.
PG5	COMMERCIAL CLOSE	98 market pigs produced (100 weaners – 2 mortality). Each pig weighed at 115 KG avg. QR code generated for each animal.	$\text{unit_cost} = \frac{\text{total_batch_cost}}{98} = \text{Rs.}(45,000 + \text{feed} + \text{OH}) / 98$ avg_weight = 115 KG/pig	Dr 1100 Inv.Pig → Cr 1190 WIP (at total batch cost / 98)	Each market pig has a precise, traceable cost from weaner to sale weight. QR on each animal enables the buyer/abattoir to trace back to the gilt batch and boar. Variances (feed, overhead) are settled. Batch CLOSED.

LOB: Goat & Sheep

Batch: BATCH-GOAT-2025-001 ! 30 Does ! Method: BIO_ASSET ! Milk + Kid + Wool output

Dairy goats are treated as Biological Assets (IAS 41), same framework as cattle. Three output types: milk (daily), kids (kidding events), and wool (shearing events for sheep).

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
GS1	PURCHASE	30 dairy goats purchased. NCA created under Biological Asset – Cattle account (same classification).	$= 30 \text{ NOS} \times \text{Rs.}8,000 = \text{Rs.}2,40,000$ NCA Same IAS 41 treatment as cattle stage = PREMATURE → MATURE on age	Dr 1220 Bio Asset Cattle → Cr 2100 AP Rs.2,40,000	Dairy goats are Balance Sheet assets. The farm's Net Asset Value immediately increases by Rs.2,40,000. Pre-mature feed and vet costs will further grow the NCA until the first milking season.
GS2	DAILY	60 litres of milk collected daily (2 L/goat/day). Unit cost computed from running NCA cost rate. Each day's collection gets a lot number = collection date.	$\text{unit_cost} = \frac{\text{daily_NCA_cost}}{60}$ lot_no = collection_date lot enables FIFO milk sale	Dr 1170 Inv.Milk → Cr 1220 Bio Asset (at running unit cost)	Goat milk often commands a premium price. Daily lot tracking enables FIFO sale of milk to cheese/dairy processors and prevents older milk being displaced by fresher stock. Cost per litre is monitored against market selling price daily.
GS3	KIDDING	Kidding event: 25 kids born (0.83 kids/doe average). Each kid is a new Biological Asset at birth value.	$= 25 \text{ NOS kids born opening NCA per kid} = \text{fair_value_at_birth}$ (appraised or std birth weight $\times$ mkt rate)	Dr 1100 Inv.Kid → Cr 1220 Bio Asset (at fair value at birth)	Kids can be: (a) retained as replacement does → new NCA batches created, (b) sold → immediate revenue, or (c) raised for meat → transferred to a standard grow-out batch. The birth event creates the asset regardless of future intent.

GS4	SHEARING	Wool harvest (applies to sheep in same LOB): 45 KG sheared. Enters inventory at NCA running rate.	= 45 KG wool entry_type = PROD_CROP unit_cost = NCA_running_rate / wool_kg	Dr 1100 Inv.Wool → Cr 1220 Bio Asset (at NCA running rate)	Wool is a secondary output of the sheep Bio Asset. Its cost basis comes from the shared NCA running cost. Farm manager can see whether wool revenue covers its allocated NCA cost or whether the flock economics depend entirely on meat/milk.
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Agriculture LOBs: Processes

LOB: Fruit Farming- Pre mature Stage

Batch: BATCH-FRUIT-PRE-2020-001 ! 100 Mango Trees ! 5 Years ! Method: BIO_ASSET ! No Output

Fruit trees are Bearer Plants under IAS 16/IAS 41. Before they reach harvest maturity (5 years for mango), All costs are capitalised into the NCA. No amortisation, no harvest, and no batch close for 5 years.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
FR1	YEAR 1 PRE- MATURE	100 mango saplings purchased and planted. NCA created. Stage = PREMATURE. System reads bio_asset_stage_config: amort_applies=FALSE, harvest_possible=FALSE.	= 100 NOS × Rs.500 = Rs.50,000 NCA stage = PREMATURE amort = FALSE harvest = FALSE cost_treatment = CAPITALISE	Dr 1240 Bearer Plant → Cr 2100 AP Rs.50,000	Trees are long-term assets. Rs.50,000 appears on the Balance Sheet. There is no P&L impact at planting. The farm must fund 5 years of nurturing costs before any revenue flows. Investors and lenders see the asset value growing over time.
FR2	YEAR 1-5 ANNUAL	Each year: fertiliser, irrigation, and labour costs totalling Rs.8,000/year are capitalised into the Bearer Plant NCA. The SAME batch record runs for all 5 years - it never closes during the premature stage.	= annual_cost = Rs.8,000 CAPITALISED into 1240 NCA NO P&L charge during premature NCA after 5 years: Rs.50,000 + 5×Rs.8,000 = Rs.90,000	Dr 1240 Bearer Plant → Cr 1100/2100 Rs.8,000/year	Over 5 years, the trees accumulate Rs.90,000 of capitalised value. Management monitors: NCA per tree (Rs.900), projected future harvests, and NPV of the orchard investment. An external valuer may assess Fair Value annually.
FR3	YEAR 5 END	Age-based trigger: trees reach 5 years (premature_years from breed_master). System auto-transitions stage to MATURE. Creates amortisation schedule. Transfers NCA within the account. Batch does NOT close.	= NCA at maturity = Rs.90,000 amort per year: (Rs.90,000 – Rs.10,000 residual) / 10 years = Rs.8,000/year monthly = Rs.667/month	Dr 1240-Mature → Cr 1240- Premature (reclassification, no P&L)	This is the critical transition. From Month 61 onward: (a) amortisation begins charging P&L Rs.667/month, (b) harvests are now possible, and (c) the annual batch copy cycle begins. The transition is automatic - no manual accounting entry needed.

LOB: Fruit Farming - Mature Stage (Annual Batch + Copy)

Batch: BATCH-FRUIT-MAT-2025-001 ! MATURE ! Annual Harvest ! FIFO lot sale

From YR 6 onward, the mango trees produce annual harvests. Each year is a separate batch (created by copying the previous year). Monthly amortisation runs throughout. Harvested mangoes enter inventory as FIFO lots.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
FM1	YEAR 6 FIRST SEASON	Annual maintenance costs (fertiliser, irrigation, pruning labour) are recorded. Unlike the pre-mature stage, these productive-stage costs accumulate in the batch as WIP costs that will determine the harvest unit cost.	= annual maintenance = Rs.8,000 entry_type = BATCH_CONSUMPTION_BIO adds to batch cost pool (separate from NCA amort)	Dr 1240 Bearer Plant → Cr 1100 Inv.Fert Rs.8,000	Management now sees two parallel P&L charges: (1) amortisation Rs.8,000/year (recovery of tree NCA cost), and (2) productive maintenance Rs.8,000/year (direct farming costs). Total annual orchard cost = Rs.16,000 before the harvest unit cost is calculated.
FM2	YEAR 6 AMORT	Monthly amortisation of Rs.667 runs automatically on the first day of each month. This reduces the NCA net book value and charges P&L.	= (Rs.90,000 – Rs.10,000) / 120 months = Rs.667/month amortisation annual = Rs.8,000/year NCA reduces each month	Dr 7100 Amort Expense → Cr 1241 Accum Amort Rs.667/month	Without amortisation, the orchard's true economic cost would be understated. The Rs.667/month flows to P&L even if no harvest occurs (For Example- drought year). This ensures the orchard investment is properly recovered over its productive life.
FM3	HARVEST SEASON	15,000 KG of mangoes harvested. Unit cost = annual NCA running cost (maintenance + amort) divided by harvest KG. NCA is NOT fully consumed - the trees continue for future years.	= unit_cost = (annual_maintenance + annual_amort) / harvest_kg = (Rs.8,000 + Rs.8,000) / 15,000 = Rs.1.07/KG (example) NCA net value continues to decline	Dr 1180 Inv.Fruit → Cr 1240 Bearer Plant 15,000 KG at unit cost	Mangoes enter inventory as FIFO Lot 1 at Rs.1.07/KG. The low cost per KG reflects the trees' long productive life spreading the NCA cost over many harvests. A bad harvest year (For Example- 8,000 KG) would produce a higher unit cost (Rs.2.00/KG) - natural cost fluctuation.
FM4	FIFO SALE	10,000 KG mangoes sold from Lot 1 (oldest harvest first). Cost is transferred to COGS at the FIFO layer rate.	= FIFO Lot 1: Rs.7.20/KG rate 10,000 KG × Rs.7.20 = Rs.72,000 COGS gross margin = (sale_price – Rs.7.20) × 10,000	Dr 5100 COGS → Cr 1180 Inv.Fruit Rs.72,000	FIFO ensures older (and often cheaper) harvest lots are sold first. If a second harvest enters inventory at Rs.8.00/KG, the first lot must be fully sold before the higher cost layer appears in COGS. This is the fruit grower's cost-accurate sales model.

FM5	YEAR END BATCH COPY	At year end, this annual batch is CLOSED. A new batch for Year 7 is created by copying the closed batch. Scheduler and location are automatically copied.	= batch_copy_log: copy_type = ANNUAL_SEASON source = BATCH-FRUIT-MAT-2025 new = BATCH-FRUIT-MAT-2026 fields_copied = {scheduler, location, breed, trees}	No journal (batch lifecycle management)	The copy mechanism eliminates manual re-setup each year. The new batch inherits the same 100 trees, same orchard location, same maintenance scheduler. Only dates are updated. The accountant simply starts adding new-year costs to the copied batch.
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LOB: Crop Farming

Batch: BATCH-CROP-2025-001 ! 10 Acres Paddy ! 120 Days ! Method: STANDARD ! QC + QR

Seasonal crop batch. Seed sowing, fertiliser, pesticide, and labour costs accumulate in WIP over 120 days. Harvest output is recorded. A favourable output variance is generated when actual yield exceeds standard. QC and QR are mandatory on harvest bags.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
CR1	DAY 1	Paddy seeds sowed: 200 KG drawn from seed inventory at Rs.45/KG. The system records this as a PROD_INPUT (the opening input of the crop batch).	= 20 KG/acre × 10 acres = 200 KG expected actual = 200 KG (matches std) cost = 200 × Rs.45 = Rs.9,000	Dr 1190 WIP → Cr 1100 Inv.Seed Rs.9,000	Seed cost is the foundation of the crop WIP. Using more seed than the standard (For Example- compensating for poor germination) would create a usage variance at close. The batch is now active for the full 120-day growing season.
CR2	DAY 30	First fertiliser application: 300 KG at Rs.25/KG. Expected = 300 KG - no deviation.	= 300 KG × Rs.25 = Rs.7,500 expected = 300 KG ✓ no variance contribution this event	Dr 1190 WIP → Cr 1100 Inv.Fert Rs.7,500	Fertiliser cost accumulates in WIP. The farm manager can see the running cost per acre at any point (currently Rs.1,650/acre after sowing + fertiliser). If a second application is needed unexpectedly, the resulting usage variance will be captured at close.
CR3	DAY 90	Pesticide spray: 20 L at Rs.120/L. Expected = 20 L - within tolerance.	= 20 L × Rs.120 = Rs.2,400 expected = 20 L ✓ no variance contribution	Dr 1190 WIP → Cr 1100 Inv.Pest Rs.2,400	Pesticide application timing (Day 90) aligns with the scheduler programme. If an unexpected pest outbreak required an extra spray, that additional cost would flag as a Usage Variance at close - drawing management attention to the event.
CR4	DAY 120	Harvest: 5,200 KG paddy actually harvested vs the standard expectation of 5,000 KG. The surplus 200 KG will generate a	= actual_harvest = 5,200 KG expected = 500 KG/acre × 10 = 5,000 KG surplus = +200 KG → FAV	Dr 1180 Inv.Crop → Cr 1190 WIP 5,200 KG at unit cost	A better-than-expected harvest reduces the cost per KG. If WIP at close = Rs.36,000, unit cost = Rs.36,000/5,200 = Rs.6.92/KG. This is lower than the Rs.6.00 standard

		Favourable Output Variance.	$\text{unit_cost} = \text{total_WIP} / 5,200$		- the extra 200 KG of yield for the same input cost is the source of the favourable variance.
CR5	CLOSE	Favourable Output Variance posted. Because actual output exceeds standard, the excess inventory value is credited to the Output Variance account (reducing expense).	$\text{var} = (\text{std_qty} - \text{actual_qty}) \times \text{std_cost} = (5,000 - 5,200) \times \text{Rs.6.00} = -\text{Rs.1,200 FAVOURABLE}$ Dr Inventory / Cr Variance	Dr 1180 Inv.Crop → Cr 6140 Output Var Rs.1,200 (FAV - Cr side)	A FAV output variance is a credit to P&L (reduction in variance expense). Management sees this as a performance reward for better-than-standard yield. The standard yield may be revised upward for next season if FAV variances repeat.
CR6	QC	Grain QC batch created for 5,200 KG of paddy. Parameters: moisture content (<13%), grain size (visual), foreign matter (<0.5%).	$\text{batch_type} = \text{QC}$ params: moisture < 13% foreign_matter < 0.5% result: moisture = 12% ✓ foreign = 0.3% ✓ overall_result = PASS	No cost journal (QC process step)	QC is mandatory before paddy can be bagged and labelled (lob_master.qc_required = YES). Failing QC (For Example- moisture >13%) would trigger QUARANTINE - the paddy must be dried before re-inspection. This protects against mould claims from buyers.
CR7	QR	5,200 KG bagged into 104 bags × 50 KG each. Each bag receives a unique QR code label.	$5,200 \text{ KG} / 50 \text{ KG} = 104 \text{ bags}$ QR payload per bag: lot, harvest_date, field_location, crop_variety, net_weight=50KG, moisture=12%, qc_result=PASS	No cost journal (QR generation)	Each 50 KG bag is now uniquely traceable. Buyers (mills, wholesalers) can scan the QR to verify the harvest date, field, and QC result. This is increasingly demanded by large buyers and export certification bodies.

LOB: Seeds- Seed Processing

Batch: BATCH-SEEDS-2025-001 ! 5,000 KG Raw Paddy In ! 4,750 KG Certified Seeds Out ! Method: STANDARD

Selected paddy from the Crop batch is processed into certified seeds. The batch_type = SEED_PROCESSING. A 5% processing loss is expected. QC (germination rate, moisture, purity) and QR on each seed bag are mandatory.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
SD1	INPUT	5,000 KG of certified-quality paddy is transferred from the Crop batch (BATCH-CROP-2025-001) into the Seed Processing WIP. Full traceability to	$5,000 \text{ KG} \times \text{Rs.6.00/KG} = \text{Rs.30,000}$ source_batch_id = BATCH-CROP-2025-001 entry_type = PROD_INPUT	Dr 1190 WIP → Cr 1180 Inv.Crop Rs.30,000	Traceability from seed to harvest: a farmer who buys these certified seeds can trace them back to the exact field and growing season. This is the basis of seed certification. The Rs.6.00/KG raw cost enters WIP as the seed's opening cost basis.

		the source field is maintained.			
SD2	PROCESSING	Seed cleaning (removing debris), grading (size sorting), and treating. Processing overhead Rs.3,000 (machinery power, labour) added to WIP.	= entry_type = OVHD_POWER processing_overhead = Rs.3,000 added to WIP cost pool	Dr 1190 WIP → Cr 2100 AP Rs.3,000	Processing cost improves the physical quality of the seed but does not change the quantity (yet - loss comes at output). The cost per KG in WIP rises from Rs.6.00 to approximately Rs.6.60 after processing overhead is added.
SD3	CHEMICAL	Seed treatment chemical applied: 10 KG at Rs.500/KG. This fungicide/insecticide coating protects seeds from soil-borne pathogens.	= 10 KG × Rs.500 = Rs.5,000 entry_type = CONS_MEDICINE item = SEED_TREATMENT	Dr 1190 WIP → Cr 1100 Inv.Chem Rs.5,000	Chemical treatment is a mandatory cost for seed certification. Without it, seeds cannot receive the certified label. The Rs.5,000 treatment cost is relatively small vs the Rs.30,000 raw paddy cost but is essential for market acceptance.
SD4	OUTPUT	4,750 KG of certified seeds produced (5% processing loss from 5,000 KG). The unit cost is computed from total WIP divided by output quantity.	= output = 5,000 × (1–5%) = 4,750 KG total_WIP = Rs.30,000 + Rs.3,000 + Rs.5,000 = Rs.38,000 unit_cost = Rs.38,000 / 4,750 = Rs.8.00/KG	Dr 1155 Inv.Seeds → Cr 1190 WIP 4,750 KG at Rs.8.00/KG = Rs.38,000	The 5% loss (250 KG) is absorbed into the unit cost - this is how the cost per output KG increases from Rs.6.00 (raw) to Rs.8.00 (certified). Certified seeds are sold at Rs.25-40/KG, producing a strong margin on the Rs.8.00 cost basis.
SD5	QC	Mandatory seed QC batch: germination test (≥85%), moisture measurement (<12%), purity test (>99% variety purity).	= QC params: germination = 88% ≥ 85% ✓ moisture = 11% < 12% ✓ purity = 99.2% > 99% ✓ overall_result = PASS	No cost journal (QC process step)	Seed QC is required by agricultural certification bodies. Failing germination (<85%) means the entire batch must be re-treated or downgraded to feed-grade paddy. The pass results are stored in qr_code_master for printing on bag labels.
SD6	QR	190 certified seed bags (25 KG each) are labelled with QR codes containing full traceability and QC results.	= 4,750 KG / 25 KG = 190 bags QR payload: crop_variety, lot, germination=88%, treatment_date, expiry_date, source_batch = BATCH-CROP-2025-001	No cost journal (QR generation)	Farmers who buy these seeds can scan the QR and see the 88% germination rate, the field of origin, and the treatment date. This transparency commands a price premium and builds brand loyalty. Expired seed bags are flagged by the system.

Sheet 11 - Aqua, Insect & Feed LOBs: Step-by-Step Process

LOB: Fish Farming

Batch: BATCH-FISH-2025-001 ! 10,000 Fingerlings ! 180 Days ! Method: BIO_ASSET + FIFO Harvest

Fish are stocked as a Bio Asset in a pond (sub-location). Feed costs are capitalised into the NCA daily. Weight observations are tracked via the KPI engine. Partial and full harvests are recorded as separate FIFO lots, enabling precise cost tracing to sale.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
AQ1	STOCKING	10,000 fingerlings stocked into Pond 1. Biological Asset NCA created. Stage = GROWING.	= 10,000 NOS × Rs.5.00 = Rs.50,000 NCA sub_location = Pond 1 bio_asset_stage = GROWING (fish don't have PREMATURE/MATURE split - GROWING from day 1)	Dr 1230 Bio Asset Fish → Cr 2100 AP Rs.50,000	The pond is now an active Bio Asset. All costs added to the NCA during the grow-out period (feed, chemicals, electricity for aeration) will eventually become the unit cost of harvested fish. Pond capacity is validated at stocking.
AQ2	DAILY FEEDING	20 KG of fish pellets fed per day. Daily cost capitalised into the fish NCA. FCR (Feed Conversion Ratio) target = 1.5 KG feed per KG weight gain.	= 20 KG × Rs.30/KG = Rs.600/day CAPITALISED into 1230 NCA running FCR = cumulative_feed / (current_biomass – opening_biomass)	Dr 1230 Bio Asset → Cr 1100 Inv.Feed Rs.600/day	Rs.600/day × 180 days = Rs.1,08,000 feed cost capitalised. Combined with Rs.50,000 stocking cost, NCA reaches Rs.1,58,000 at harvest. Unit cost per KG of fish = Rs.1,58,000 / total harvest KG. Monitoring FCR daily catches overfeeding before costs balloon.
AQ3	WEEK 4	Sample weight observation: 5 fish sampled, average 45 grams. KPI expects 50 grams at Week 4 based on breed_master benchmark.	= entry_type = DESC_WEIGHT actual = 45g kpi_target = 50g at week 4 deviation = –10% below target → WARNING alert fired	No journal (descriptive observation)	A 10% weight lag at Week 4 is a critical early warning. Causes investigated: feed quality, dissolved oxygen levels, stocking density, or disease. If not corrected, the fish will be underweight at harvest, reducing revenue and increasing unit cost per KG. Alert goes to Farm Manager immediately.
AQ4	MONTH 4 PARTIAL	Partial harvest: 2,000 KG of larger fish selectively harvested from the pond. This creates FIFO Lot 1. Unit cost = NCA running balance / total estimated biomass.	= partial_harvest = 2,000 KG unit_cost = NCA_running / total_est_biomass FIFO Lot 1 created @ this unit_cost NCA reduces by harvested portion	Dr 1140 Inv.Fish → Cr 1230 Bio Asset 2,000 KG at unit cost	Partial harvests allow the farm to supply premium-size fish to market while allowing smaller fish to continue growing. FIFO Lot 1 has a lower unit cost (early harvest, less feed accumulated). This is sold at a higher price margin.

AQ5	MONTH 6 FULL	Final harvest: 5,800 KG. NCA reaches zero. Batch COMPLETED and CLOSED. FIFO Lot 2 created at the final unit cost.	= final_harvest = 5,800 KG NCA → 0 (fully consumed) FIFO Lot 2 @ final unit_cost total harvested = 2,000 + 5,800 = 7,800 KG batch status → CLOSED	Dr 1140 Inv.Fish → Cr 1230 Bio Asset 5,800 KG at unit cost NCA = 0	Unlike fruit trees, fish ponds do NOT use the annual copy mechanism - a new stocking batch is created from scratch each cycle. The two FIFO lots (Lot 1 and Lot 2) will be sold in order. If Lot 2 was harvested later, its cost per KG is slightly higher due to additional feed.
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LOB: Aqua Slaughter

Batch: BATCH-AQSL-2025-001 ! 7,800 KG Live Fish In ! QC + QR + Joint Cost Split

Live fish from the Fish Farming batch enter the processing facility. Cost split: 70% Fillet, 15% Fish Meal, 10% Fish Skin, 5% Waste. QC checks freshness and fillet grade. QR code on each pack.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
AS1	INPUT	7,800 KG live fish transferred from Fish inventory into Slaughter WIP. Cost is carried at the FIFO rate from the fish farming batch.	= 7,800 KG × Rs.14.50/KG (FIFO rate) = Rs.1,13,100 enters WIP source_batch_id = BATCH-FISH-2025-001	Dr 1190 WIP → Cr 1140 Inv.Fish Rs.1,13,100	The entire fish farming cost (stocking + 6 months feed + overheads) is now in Slaughter WIP. This single figure represents the full farm-gate cost before processing. The slaughter batch will add processing cost on top and then split the total across output products.
AS2	PROCESS	Slaughter processing overhead: labour (filleting team), ice (chilling), packaging materials. Rs.12,000 added to WIP.	= total_WIP = Rs.1,13,100 + Rs.12,000 = Rs.1,25,100 before split entry_type = OVHD_LABOR	Dr 1190 WIP → Cr 2100 AP Rs.12,000	Processing cost must be added to WIP before any split. Otherwise, by-products would not carry their fair share. Rs.12,000 processing on Rs.1,13,100 input = Rs.1.54/KG additional cost. Management monitors labour productivity: KG processed per man-hour.
AS3	OUTPUT 1	Main product: Fish Fillet. 5,460 KG produced = 70% of input weight. 70% of total WIP cost allocated.	= split = 70% allocated = Rs.1,25,100 × 0.70 = Rs.87,570 unit_cost = Rs.87,570 / 5,460 = Rs.16.04/KG	Dr 1140 Inv.Fillet → Cr 1190 WIP Rs.87,570	Fish fillet at Rs.16.04/KG enters inventory. Typical retail price Rs.80-120/KG - strong margin. The unit cost feeds into COGS when fillet is sold. QC and QR are completed before the fillet leaves the facility.
AS4	OUTPUT 2	By-product: Fish Meal. 1,170 KG = 15% cost split.	= Rs.1,25,100 × 0.15 = Rs.18,765 unit_cost = Rs.18,765 / 1,170 = Rs.16.04/KG	Dr 1100 Inv.Meal → Cr 1190 WIP Rs.18,765	Fish meal is sold to feed manufacturers at Rs.30-40/KG. At Rs.16.04/KG cost, this by-product is profitable. The 15% split is based on the historical yield of this fish

					species and can be adjusted in slaughter_cost_split_config.
AS5	OUTPUT 3	By-product: Fish Skin. 780 KG = 10% cost split.	= Rs.1,25,100 × 0.10 = Rs.12,510 unit_cost = Rs.12,510 / 780 = Rs.16.04/KG	Dr 1100 Inv.Skin → Cr 1190 WIP Rs.12,510	Fish skin is used in collagen products and pet food. At Rs.16.04/KG cost and market price Rs.20-30/KG, it is marginally profitable. The remaining 5% of cost goes to Waste/Stock Adjustment Loss, closing WIP to zero.
AS6	QC + QR	QC batch created: 5,460 KG fillet inspected for freshness score (1-5 scale) and fillet grade (A/B/C). After PASS, QR codes printed on each pack.	= QC params: freshness_score ≥ 3 of 5 fillet_grade = A or B QR payload: harvest_date, pond_id, weight, fillet_grade, lot_no, expiry	No cost journal (QC/QR process steps)	Export-grade fish fillet (Grade A, freshness 5/5) commands a premium export price. The QR code is scanned at the port of exit and by the importing country for food safety compliance. Grade C fillets are redirected to the domestic market or further processed.

LOB: Bee Keeping

Batch: BATCH-BEE-2025-001 ! 50 Hives ! Annual ! Method: STANDARD ! QR on Honey Jars

50 beehives produce honey over an annual cycle. Monthly feed (sugar) and labour costs accumulate in WIP. One annual harvest produces the main product (honey) and by-product (beeswax). QC checks moisture and purity. 1,750 jars receive QR codes.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
BK1	SETUP	50 hives purchased and installed at Rs.3,000/hive. Hives are recorded as fixed assets (not NCA bio assets). Bees may optionally be treated as a bio asset.	= 50 hives × Rs.3,000 = Rs.1,50,000 treated as FIXED ASSET (not NCA - hives are equipment that contains living bees) honey output → STANDARD costing	Dr Fixed Asset A/c → Cr 2100 AP Rs.1,50,000	This is a pragmatic accounting choice: hives are equipment, honey is standard-costed output. The farm can optionally upgrade bees to BIO_ASSET if hive level NCA tracking is needed. For most small-to-medium apiaries, STANDARD costing is sufficient.
BK2	MONTHLY	Monthly sugar feed: 100 KG × Rs.40/KG. Bees are fed sugar syrup during low-nectar seasons to maintain colony health.	= 100 KG × Rs.40 = Rs.4,000/month × 12 months = Rs.48,000/year entry_type = CONS_FEED	Dr 1190 WIP → Cr 1100 Inv.Feed Rs.4,000/month	Sugar feed is the primary variable cost for apiary operations. In good nectar seasons, sugar consumption drops and WIP cost is lower - yielding a better honey unit cost. In drought/winter, more sugar is fed and unit cost rises. Monthly WIP tracking shows this dynamic.

BK3	MONTHLY	Labour and hive maintenance: Rs.2,000/month for the beekeeper's time and consumables (wax foundation, hive frames).	= Rs.2,000/month overhead × 12 months = Rs.24,000/year added to WIP	Dr 1190 WIP → Cr 2100 AP Rs.2,000/month	The beekeeper's labour is the second major cost. If the same person manages multiple apiaries, time must be allocated across batches. Resource cost tracking enables management to determine the optimal hive-to-beekeeper ratio.
BK4	HARVEST (June)	Annual honey harvest: 875 KG raw honey extracted. Unit cost = total batch WIP / 875 KG.	= total_WIP = Rs.48,000 feed + Rs.24,000 labour = Rs.72,000 (annual) unit_cost = Rs.72,000 / 875 = Rs.82.29/KG (before beeswax by-product split)	Dr 1160 Inv.Honey → Cr 1190 WIP 875 KG at unit cost	Raw honey at Rs.82.29/KG enters inventory. After QC and jarring, the final unit cost per 500g jar will be higher (packaging + QC cost added). Retail price Rs.500-800 per 500g jar - excellent margin at Rs.41.15 raw cost per 500g.
BK5	BY-PRODUCT	45 KG beeswax harvested alongside honey. 5% of batch cost is allocated to beeswax.	= beeswax cost = total_WIP × 5% = Rs.72,000 × 0.05 = Rs.3,600 unit_cost = Rs.3,600 / 45 = Rs.80/KG honey gets remaining 95% of cost	Dr 1160 Inv.Beeswax → Cr 1190 WIP Rs.3,600 (5% split)	Beeswax has a stable industrial demand (cosmetics, candles, polish). At Rs.80/KG cost and market price Rs.200-300/KG, beeswax is a profitable by-product. The 5% cost allocation is configurable and can be adjusted based on actual beeswax value.
BK6	QC	Honey QC: moisture percentage, colour grade, and HMF (Hydroxymethylfurfural - a honey quality indicator) are measured.	= QC params: moisture = 16% < 18% ✓ HMF = 35 mg/kg < 40 mg/kg ✓ colour = Grade 1 (light amber) overall_result = PASS	No cost journal (QC process step)	International honey standards (Codex Alimentarius) require moisture <20% and HMF <80 mg/kg. Premium markets (EU export) require moisture <18% and HMF <40 mg/kg. Passing QC at these thresholds opens export market access at Rs.600-800/kg wholesale.
BK7	QR	875 KG honey bottled into 1,750 jars × 500g. Each jar receives a QR code.	= 875,000g / 500g = 1,750 jars QR payload: harvest_date, hive_location, variety (multifloral/monofloral), moisture=16%, grade, lot_no	No cost journal (QR generation)	QR-labelled honey jars command a 15-20% premium in retail and e-commerce. The QR gives consumers confidence in the source, freshness, and purity. Organic/certified apiaries can include certification details in the QR payload.

LOB: Feed Production (Bill of Recipe)

Batch: BATCH-FEED-2025-001 ! 1,000 KG Layer Mash ! Method: STANDARD ! BOR-2025-001 V1

The feed mill produces compound feed using a Bill of Recipe (BOR). Raw materials are issued per BOR inclusion percentages. Resource and indirect costs are added. Output feed is QC tested. A Usage Variance fires if actual quantities differ from BOR quantities.

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
FP1	BATCH SETUP	User creates the batch and attaches BOR-2025-001 Version 1. The system automatically copies all BOR ingredient lines, resource lines, and indirect cost lines into batch_input_lines. Version 1 is now locked in this batch.	= bor_id = uuid-bor-001 version_no = 1 system copies BOR lines → batch_input_lines (auto-populated) BOR version locked: cannot change after APPROVE	No journal (batch setup)	Auto-population from BOR eliminates manual re-entry of the recipe. If a nutritionist updates the BOR to Version 2 (different recipe), this batch remains on Version 1. Future batches can choose V2. Version control ensures each batch's recipe is auditable.
FP2	ISSUE	Maize issued from Raw Material inventory to the batch: 520 KG = 52% inclusion × 1,000 KG batch size.	= BOR: inclusion = 52% qty = 52% × 1,000 = 520 KG actual = 520 KG (matches BOR) cost = 520 × Rs.0.35 = Rs.182	Dr 1190 WIP → Cr 1100 Inv.RM Rs.182	The system validates that the actual issued quantity matches the BOR quantity. If the miller issues 530 KG (10 KG extra), a Usage Variance will be created at close. The BOR inclusion percentage and batch size together determine the expected quantity - no manual calculation needed.
FP3	ISSUE	Soybean Meal: 300 KG issued (30% × 1,000 KG). The protein contribution from soybean is the largest in the recipe.	= BOR: inclusion = 30% qty = 300 KG cost = 300 × Rs.0.62 = Rs.186 protein_contribution: 30% × 46% protein = 13.80%	Dr 1190 WIP → Cr 1100 Inv.RM Rs.186	The system tracks both the cost and the nutritional contribution of each issued ingredient against the BOR targets. If soybean protein drops (different supplier), the nutritionist is alerted that actual protein% may fall below the 18% target specification.
FP4	ISSUE	Rice Bran: 50 KG issued (5% × 1,000 KG). Lower-cost energy source in the recipe.	= qty = 5% × 1,000 = 50 KG cost = 50 × Rs.0.25 = Rs.12.50	Dr 1190 WIP → Cr 1100 Inv.RM Rs.12.50	Rice bran is a cost-reduction ingredient. If the nutritionist wants to increase rice bran and reduce the more expensive soybean, they create BOR Version 2. This batch remains on V1 - the cost

					difference becomes visible in the version comparison report.
FP5	ISSUE	Micro-ingredients issued: Limestone (1%) + DCP (2%) + Oil (3%) + Premix (2%) = remaining 8%. Combined cost Rs.88.	= Limestone: 10KG×Rs.0.10 = Rs.1 DCP: 20KG×Rs.0.55 = Rs.11 Oil: 30KG×Rs.1.20 = Rs.36 Premix: 20KG×Rs.2.00 = Rs.40 total micro = Rs.88	Dr 1190 WIP → Cr 1100 Inv.RM Rs.88 (combined)	Micro-ingredients are small in quantity but critical for nutritional balance (calcium from limestone, phosphorus from DCP, vitamins from premix). Their cost is precisely tracked per BOR line. Running total RM cost = Rs.468.50 for 950 KG of raw materials.
FP6	RESOURCE	Labour (Rs.25) + Electricity (Rs.15) + Steam (Rs.10) charged to WIP from the BOR resource lines.	= Labour: 25 hrs × Rs.1.00 = Rs.25 Electricity: 15 kWh × Rs.1.00 = Rs.15 Steam: 10 units × Rs.1.00 = Rs.10 total resources = Rs.50	Dr 1190 WIP → Cr 2100 AP Rs.50	Resource costs flow from the BOR's resource lines directly into WIP. If actual labour hours exceed BOR hours (pelletiser breakdown causing overtime), the excess creates a Usage Variance on the resource parameter at close.
FP7	INDIRECT	Packing cost Rs.12 (BOR indirect cost line). This covers the cost of feed bags, printing, and labelling.	= packing = Rs.12 total_WIP now: RM Rs.468.50 + Resources Rs.50 + Indirect Rs.12 = Rs.530.50	Dr 1190 WIP → Cr 2100 AP Rs.12	All three cost components are now in WIP: raw materials (88.3%), resources (9.4%), indirect (2.3%). The total Rs.530.50 for 1,000 KG = Rs.0.5305/KG finished feed. This matches the BOR's theoretical cost/KG - a zero-variance scenario if all actuals matched BOR.
FP8	OUTPUT	1,000 KG of finished Layer Mash produced and transferred to Finished Feed Inventory.	= output = 1,000 KG unit_cost = Rs.530.50 / 1,000 = Rs.0.5305/KG = Rs.530.50/tonne	Dr 1150 Inv.Feed → Cr 1190 WIP 1,000 KG at Rs.0.5305/KG	Finished feed enters inventory at Rs.530.50/tonne. This is the cost that will flow into WIP when this feed is consumed by a Laying or CB batch. If the market price of Layer Mash is Rs.700/tonne, the feed mill earns a Rs.169.50/tonne margin.
FP9	QC	Feed QC: protein% (≥18%), moisture (<12%), pellet hardness. All are PASS.	= QC params: protein = 18.2% ≥ 18% ✓ moisture = 11% < 12% ✓ pellet_hardness = PASS ✓ overall_result = PASS	No cost journal (QC process step)	Feed QC is critical because this feed will go into animal batches. Substandard feed (low protein, high moisture causing mould) would degrade animal performance and trigger cascade variances across all downstream

					batches. QC at the feed mill is the first line of defence.
FP10	VARIANCE	If actual soybean issued was 310 KG (10 KG over BOR 300 KG), a Usage Variance is created at close. The actual nutritional profile is also compared to BOR.	= Usage Var = (actual – BOR_qty) × std_rate = 10 KG extra soybean × Rs.0.62 = Rs.6.20 PLANNED nutritional: actual protein 18.9% vs BOR target 18.82% → ✓ within spec	Dr 6120 Usage Var → Cr 1190 WIP Rs.6.20 (PLANNED)	Rs.6.20 variance is small but the nutritional comparison is important: if actual protein% exceeds BOR by more than 1%, it signals over-inclusion of expensive ingredients. BOR nutritional impact report gives the nutritionist visibility into recipe deviation vs specification.

Variance Engine: Functional details:

Variance Engine runs automatically when a STANDARD costing batch changes status to CLOSED. It calculates four variance types, creates journal entries for each, and validates that total debits = total credits before posting. If the journal does not balance, the system rejects the close and alerts the user.

Variance Calculation Steps - CB Farming Example

Batch: BATCH-CB-2025-001 ! STANDARD ! 10,000 DOC ! 42 Days

Step	Phase	What the System Does	Calculation / Rule	GL Entry (Dr → Cr)	Business Impact & Outcome
V1	TRIGGER	User changes batch status to CLOSED. System detects resolved_costing_method = STANDARD and triggers the variance engine. All four variance calculations run in sequence.	= TRIGGER: batch.status → CLOSED AND resolved_costing_method = 'STANDARD' → run variance_engine(batch_id)	No journal yet (trigger only)	The variance engine is fully automatic - no user input required after status change. If the engine finds data missing (For Example- no actual medicine doses recorded), it alerts the user to complete the data before variance can be finalised.
V2	PRICE VARIANCE	Compares the actual feed purchase price vs the locked standard price. Any price difference × actual quantity consumed = Price Variance.	= (actual_rate – std_rate) × actual_qty_consumed = (Rs.21.50 – Rs.20.00) × 86,400 KG = Rs.1.50 × 86,400 = Rs.1,29,600 PLANNED	Dr 6100 Price Var → Cr 1100 Inv.Feed Rs.1,29,600	Price variance is procurement's accountability. Rs.1,29,600 PLANNED on this single batch signals that feed was bought Rs.1.50/KG above standard. If this persists, procurement must re-negotiate contracts or the standard cost must

					be updated for future batches.
V3	USAGE VARIANCE	Compares actual feed consumed vs the standard quantity expected (std_rate × batch_days × actual head count). The difference × standard price = Usage Variance.	$\text{std_qty} = 0.200 \text{ KG/bird/day} \times 9,990 \text{ avg birds} \times 42 \text{ days} = 83,916 \text{ KG expected}$ $\text{actual} = 86,400 \text{ KG}$ $\text{extra} = 2,484 \text{ KG} \times \text{Rs.20.00} = \text{Rs.49,680 PLANNED}$	Dr 6120 Usage Var → Cr 1190 WIP Rs.49,680	Usage variance is farm operations' accountability. 2,484 KG of excess feed per batch is significant - equivalent to 1.5 extra days of feeding. Root causes: feed spillage, inaccurate weighing, or over-feeding. Persistently PLANNED usage variance triggers FCR review.
V4	MEDICINE USAGE	Compares actual medicine doses administered vs standard doses scheduled in the batch programme.	$\text{std_doses} = 10,000 \text{ (ND vaccine)}$ $\text{actual_doses} = 9,992$ $\text{var} = (9,992 - 10,000) \times \text{Rs.15} = -8 \times \text{Rs.15} = -\text{Rs.120 FAV}$ (8 fewer doses used)	Dr 1190 WIP → Cr 6130 Med Var Rs.120 (FAV - reversed direction)	FAV medicine variance means 8 fewer doses than scheduled were used. This could mean: fewer live birds at vaccination day (mortality reduced the count) or incomplete vaccination (a biosecurity risk). Management investigates cause before treating as a saving.
V5	OUTPUT VARIANCE	Compares standard expected output (opening qty × expected survival rate) vs actual output birds.	$\text{expected} = 10,000 \times 98\% = 9,800 \text{ birds}$ $\text{actual} = 9,750 \text{ birds}$ $\text{var} = (9,800 - 9,750) \times \text{Rs.85} = 50 \times \text{Rs.85} = \text{Rs.4,250 PLANNED}$	Dr 6140 Output Var → Cr 1190 WIP Rs.4,250	Output variance captures mortality above the standard expectation. 50 extra deaths (2.5% vs 2% expected) × Rs.85 standard cost = Rs.4,250 write-off beyond expectation. Cumulative output variance across the production chain shows the true cost of above-normal mortality.
V6	OVERHEAD VARIANCE	Compares actual overhead spend vs the standard overhead absorbed based on actual output.	$\text{std_absorbed} = \text{Rs.5.00/bird} \times 9,750 \text{ actual output} = \text{Rs.48,750 absorbed}$ $\text{actual_OH} = \text{Rs.55,000}$ $\text{OH_var} = \text{Rs.55,000} - \text{Rs.48,750} = \text{Rs.6,250 PLANNED}$	Dr 6150 OH Var → Cr 1190 WIP Rs.6,250	Overhead variance can have two causes: (a) spending more than budgeted on overheads (spending variance), or (b) producing fewer units to absorb the fixed

					overhead (volume variance). Here the lower-than-expected output means overhead is under-absorbed.
V7	BALANCE CHECK	System sums all debit amounts and all credit amounts. If DR $\neq$ CR, the journal is REJECTED and the batch cannot close.	= total_DR = sum of all debit variance journal lines total_CR = sum of all credit lines balanced = (total_DR == total_CR) IF NOT balanced $\rightarrow$ REJECT, alert user	System validation (no posting if unbalanced)	This is the fundamental double-entry safeguard. A rejected close prompts the user to review all entries - typically a missing entry or rounding difference. The system cannot post an unbalanced journal under any circumstance.
V8	POSTING & WIP CLOSE	All variance journals are posted. WIP account balance is brought to zero. Batch costing_status changes to FINALIZED.	= WIP_balance_after_variances should = 0 costing_status $\rightarrow$ FINALIZED batch_status $\rightarrow$ CLOSED	All 4 variance journals posted atomically	WIP at zero confirms complete cost settlement. The batch is now fully auditable: every rupee that entered WIP (inputs, overheads, resources) has either exited as output inventory or been recognised as a variance in P&L. Accountants can produce accurate COGS reports.

Variance Summary- Accountable

Variance Type	Accountable Party	Favourable Signal	Planned/unplanned Signal	Action If Persistently planned
Price Variance	Procurement Manager	Bought feed below standard price	Feed price above standard; supplier rate increase	Re negotiate contracts; update std cost for next cycle
Usage Variance	Farm Operations Manager	Birds consumed less feed than scheduled	FCR problem, feed wastage; inaccurate recording	Audit daily feeding records; check feeder calibration, vet check
Medicine Usage Variance	Farm / Biosecurity Manager	Fewer doses used (usually means fewer live birds)	More doses than scheduled	Investigate: disease outbreak requiring extra treatment vs inaccurate data entry

Output Variance	Farm Manager / Veterinarian	Higher survival / more birds than expected	Above-standard mortality; yield loss	Vet investigation; post-mortem analysis; review vaccination schedule
Overhead Variance	Operations / Finance	Actual overhead below budget	Over-spend on utilities, labour, repairs	Cost-centre review; maintenance audit; utility contract renegotiation

Summary: End to End Data Flow

Below diagram below shows how all LOBs connect in a complete production chain, with cost and traceability flowing through each step.

Complete Poultry Chain - Cost & Traceability Flow

Breeder Flock (Bio Asset NCA) → produces Hatching Eggs
Hatching Eggs → Hatching Batch → Day-Old Chicks (DOC) at Rs.3.97/chick
DOC → Rearing Batch → Grower Chicks at Rs.23.28/chick
Grower Chicks → CB Farming Batch → Live Birds at Rs.53.21/bird
Live Birds → Slaughter Batch → Carcass at Rs.35.71/KG (75% split) + Offal + Feather + Waste
Carcass → QC Batch (9,500 PASS) → QR Code per Pack → Consumer
At every arrow: batch_input_lines.source_batch_id creates the traceable link

Complete Agri Chain - Fruit Tree Cost Flow

Year 1: Plant 100 trees @ Rs.500 each = Rs.50,000 NCA (PREMATURE stage)
Years 1-5: Capitalise Rs.8,000/year maintenance → NCA grows to Rs.90,000
Year 5 end: Stage → MATURE. Amortisation schedule: Rs.667/month begins
Year 6+: Annual harvest. unit_cost = (Rs.8,000 maint + Rs.8,000 amort) / 15,000 KG = Rs.1.07/KG
Annual batch CLOSE → COPY batch for next season (scheduler + location auto-copied)
Harvest enters FIFO inventory → oldest lot sold first

Key Design Rules for All LOBs

1. STANDARD batches: 4 variances auto-calculated on CLOSE. WIP must zero out.
2. FIFO batches: no variance entries. FIFO layers track true cost. WIP zeroes at close.
3. BIO_ASSET batches: costs capitalise to NCA. Amort charges P&L. Harvest transfers NCA→Inventory.
4. Every inventory movement creates a balanced double-entry journal (auto-generated).
5. Every batch input line has a source_batch_id - this is the backbone of farm-to-fork traceability.
6. QC is mandatory for selected LOBs before output can leave the facility.
7. QR codes are generated ONLY after QC PASS and encode the full traceability chain.